Recurrent Neural Networks for NLP

Intro to AI - Spring 2025 Class

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1. What is a neural network?

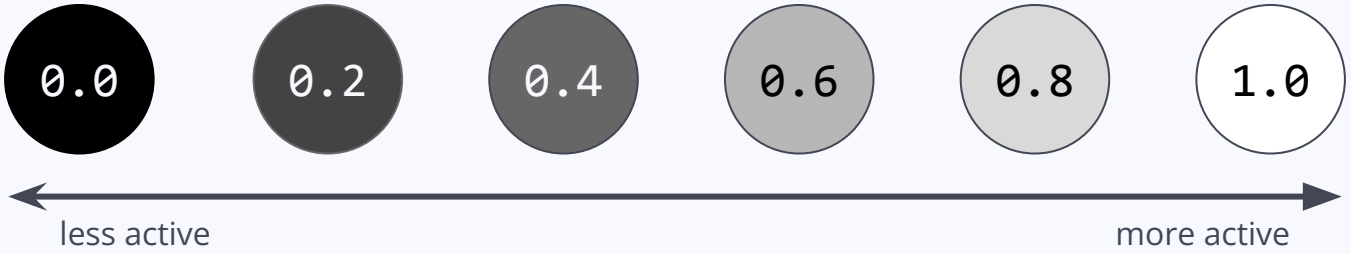
Neural Networks

- Machine learning algorithm
- Learning **features** directly from data
- Types of neural networks
  - Convolutional Neural Networks (CNNs) → great for image processing
  - Recurrent Neural Networks (RNNs, LSTMs) → great for sequential data
  - Transformers → great for… everything!

Neural Networks: Neuron

- **Neuron**: the building block of a neural network

  (other names: **node, cell**)
- Stores information, is connected to other neurons



https://www.3blue1brown.com/lessons/neural-networks

Neural Networks: Neuron

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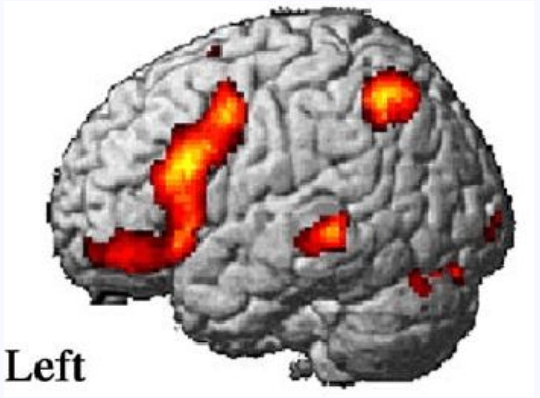
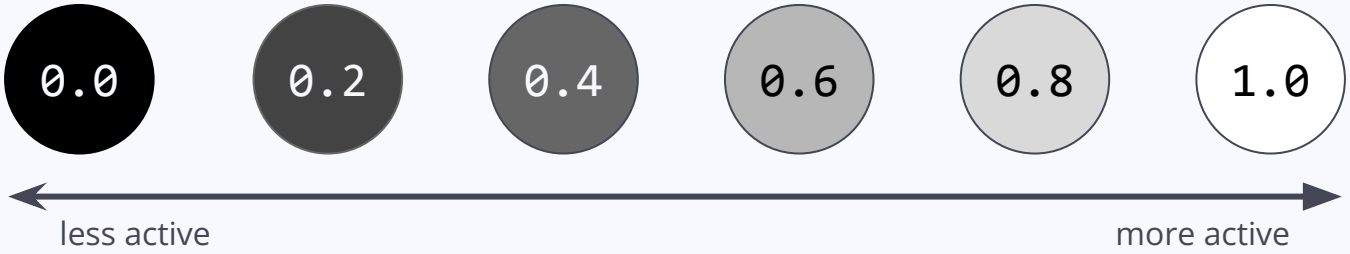


Yokoyama, Satoru, et al. "Cortical activation in the processing of passive sentences in L1 and L2: An fMRI study." Neuroimage 30.2 (2006): 570-579.



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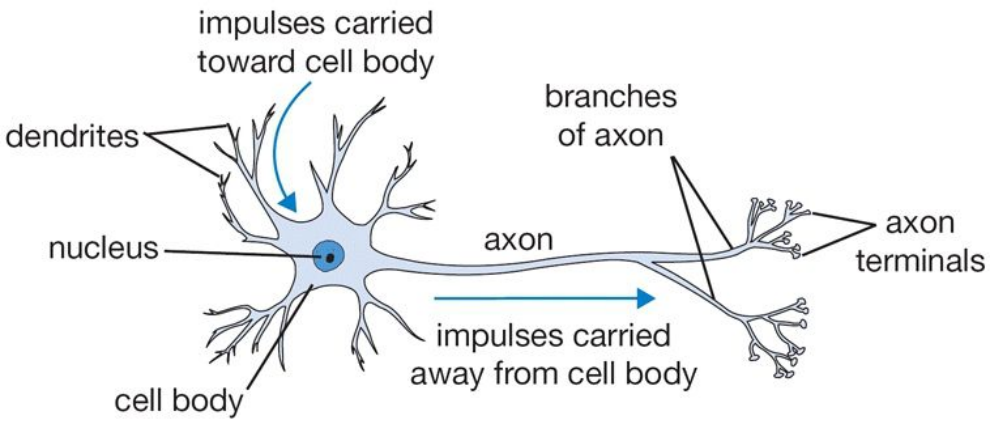
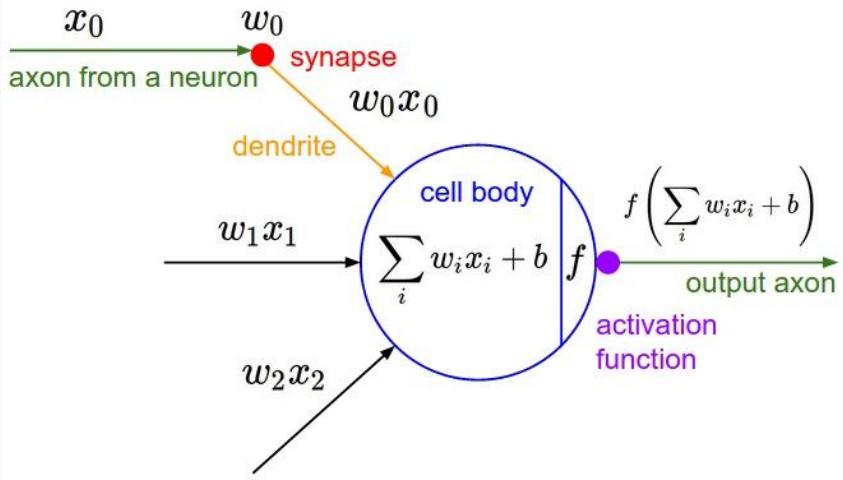
Neural Networks: Neuron



Biological neuron



Artificial neuron

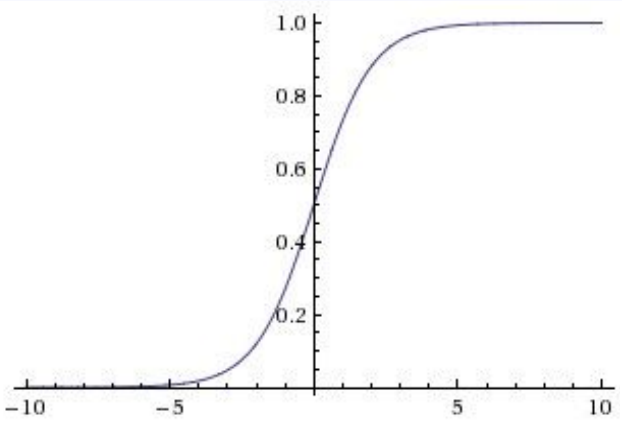
https://cs231n.github.io/neural-networks-1/#quick

Neural Networks: Neuron

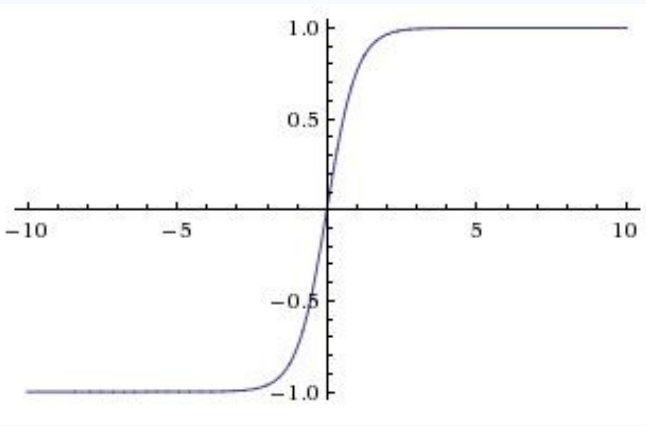
Inside a neuron...

- Dot product of **input** and **weights**: the connection between two neurons
- Add the **bias**: a value that influences the importance of the neuron
- Apply **activation function**: the *firing rate* of the neuron

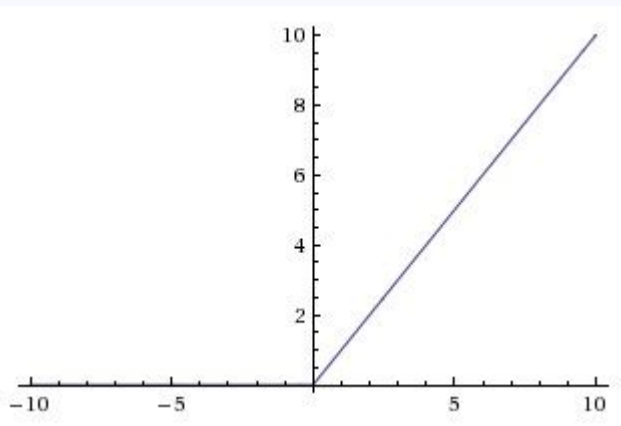
Neural Networks: Activation functions



sigmoid (-1 - 1)



tahn (-1 - 1)



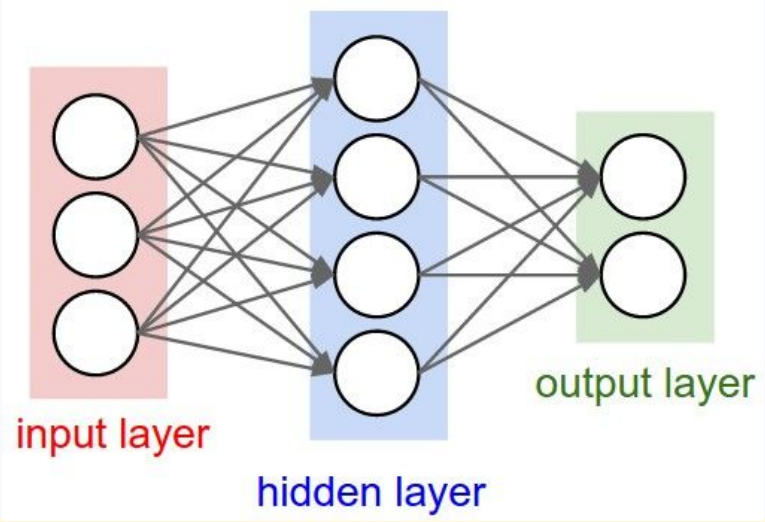
ReLU (0 - 1)

See more activation functions:
https://cs231n.github.io/neural-networks-1
https://sebastianraschka.com/faq/docs/activation-functions.html

Neural Networks: Layer

- One neuron alone? A **binary classifier**! (See Lecture 5)
- **Layer**: a collection of neurons
- **The outputs of neurons become inputs to other neurons**

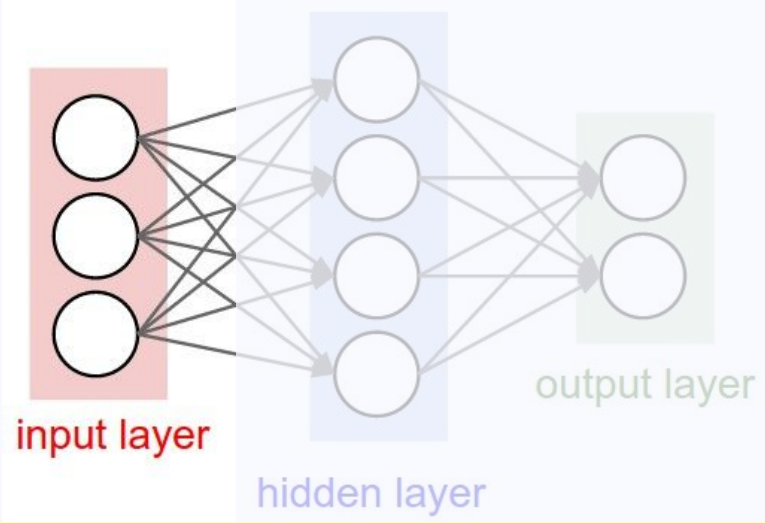
neurons of the same layer are **not** connected to each other...



...but each neuron is connected to **every** neuron of the next layer = fully-connected layer

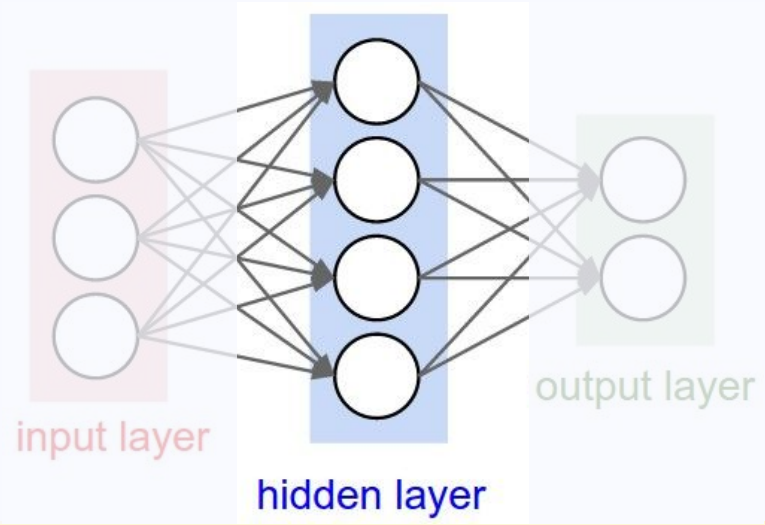
Neural Network: Layer

- **Input layer**: first layer of the NN, contains **encoded data**



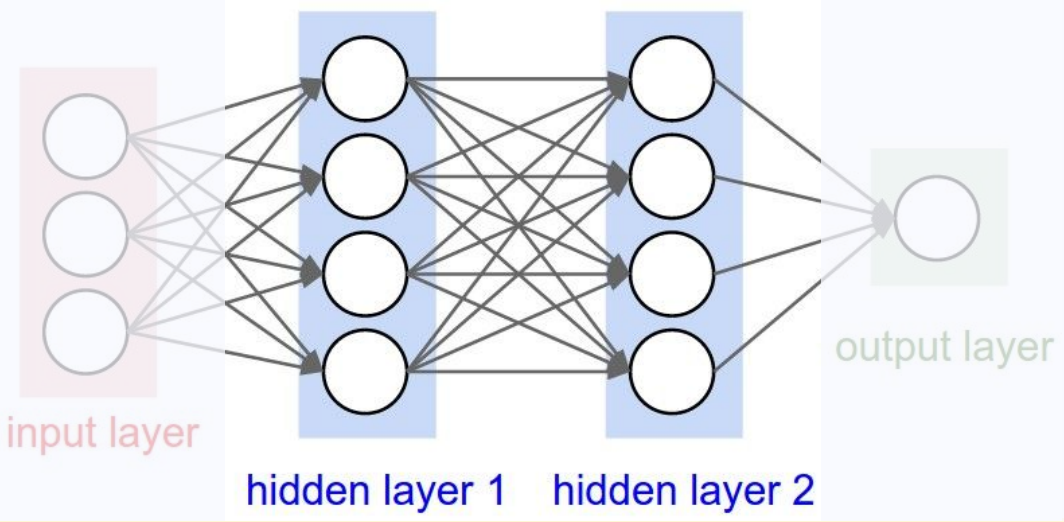
Neural Network: Layer

- **Hidden layer(s)**: learn different aspects about the data by minimizing an error/cost function in each node



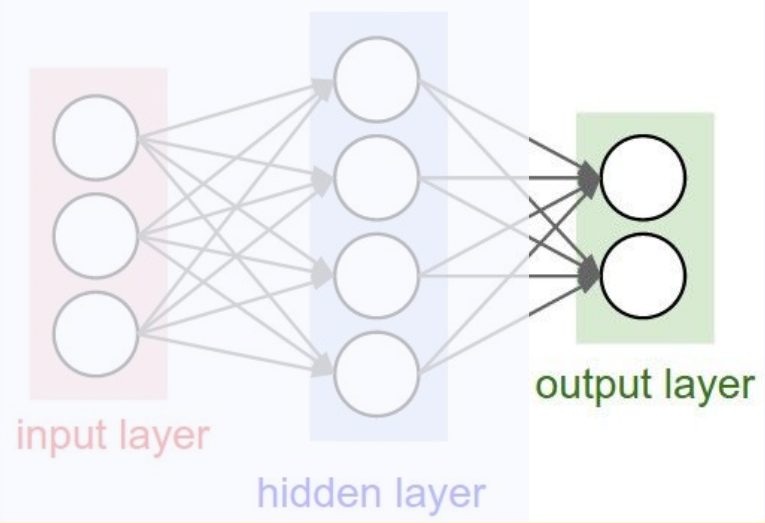
Neural Network: Layer

- **Hidden layer(s)**: learn different aspects about the data by minimizing an error/cost function in each node
- One or more hidden layers, interacting with the next!



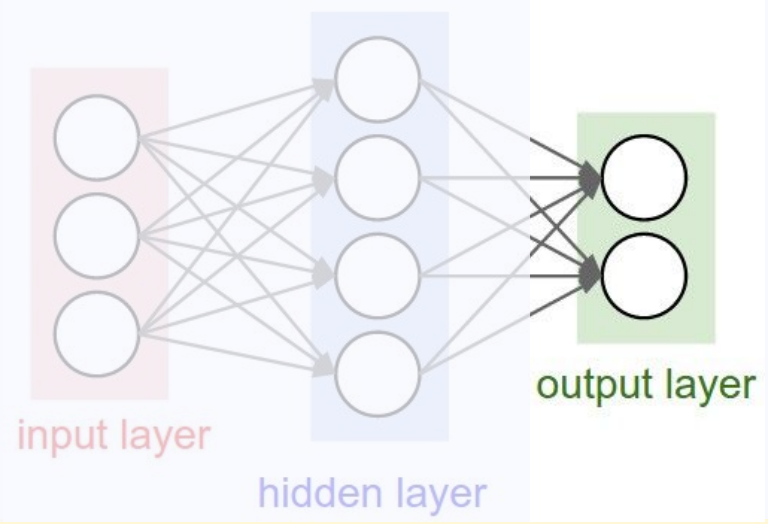
Neural Network: Layer

- **Output layer**: no function, contains values (probability of a token, label...)



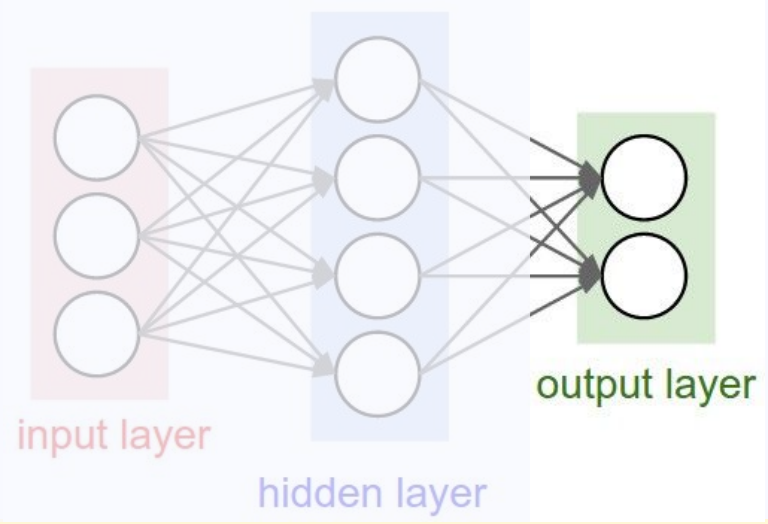
Neural Network: Layer

- **Output layer**: no function, contains values (probability of a token, label...)
- **Loss**: compare input values to predictions (= output values)



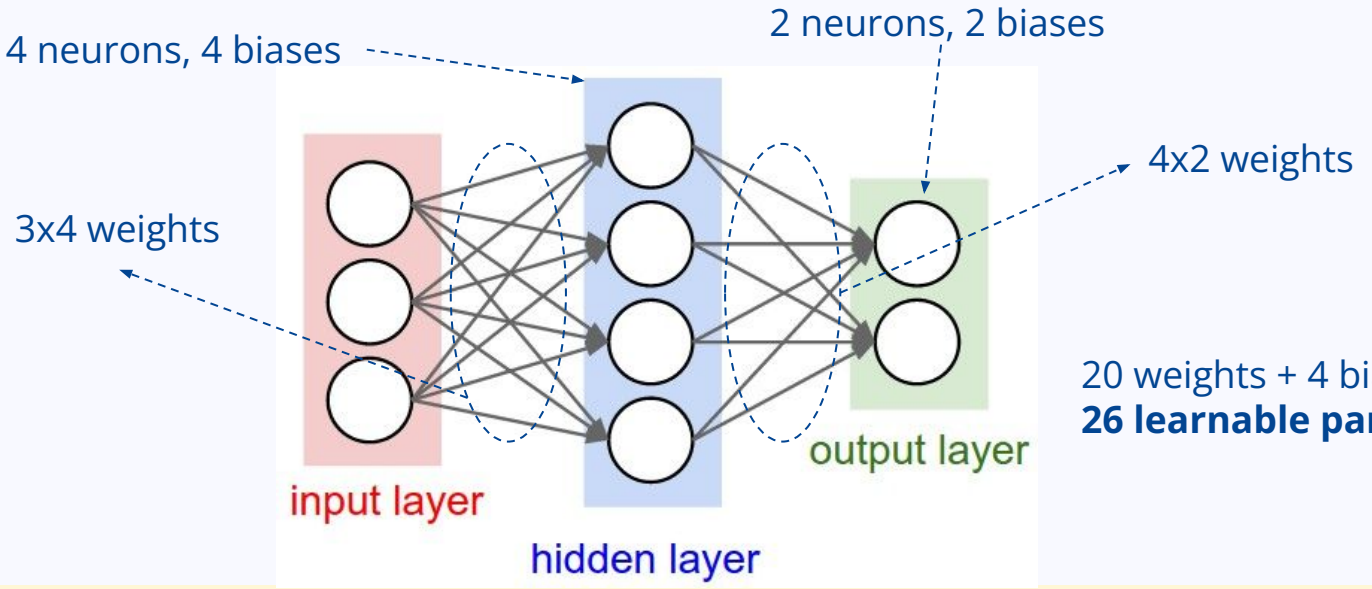
Neural Network: Layer

- **Output layer**: no function, contains values (probability of a token, label…)
- **Loss**: compare input values to predictions (= output values)
- If it's bad… re-adjust weights & biases in a new **epoch**!



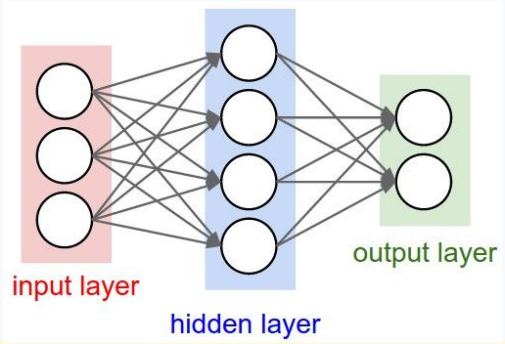
Neural Network: Size

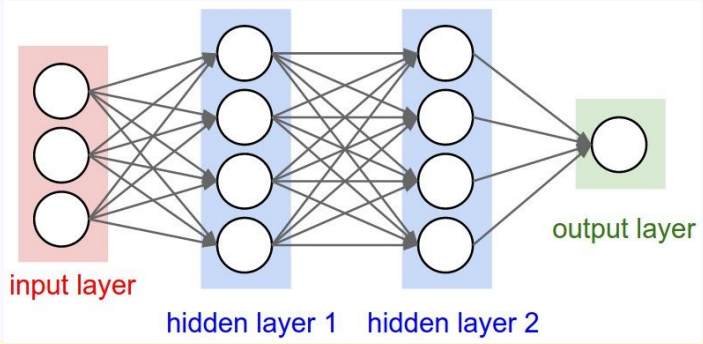
- Size based on **neurons** = Count of neurons of hidden layer(s) & output layer
- Size based on **learnable parameters** = Their biases + Their weights



Neural Network: Feedforward NN

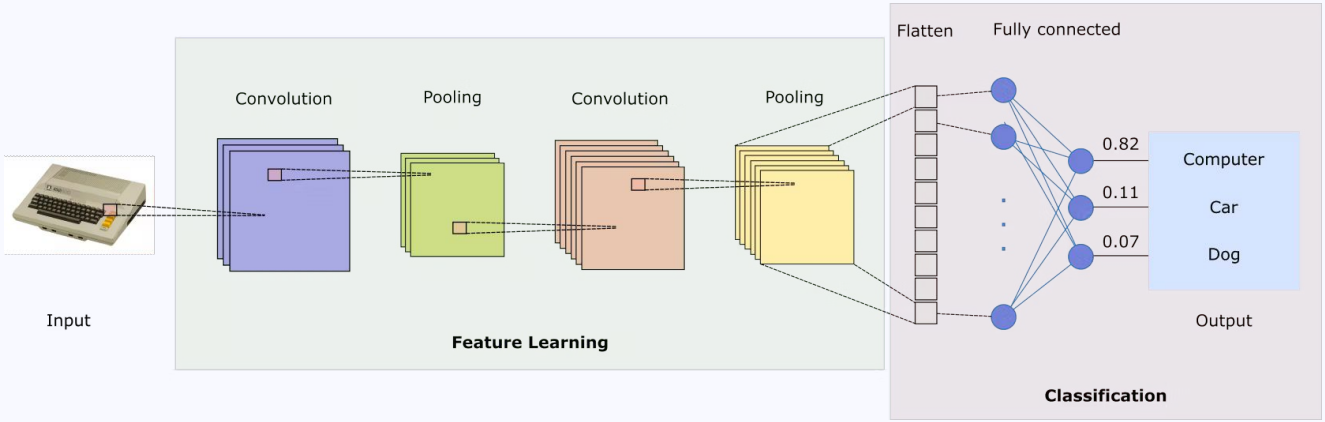
- Information flows in one direction
- (Some) drawbacks:
  - Information flows in one direction → limited contextual understanding!
  - Overfitting: good with seen data, bad with unseen data
  - Requires feature engineering (see Lecture 5!)





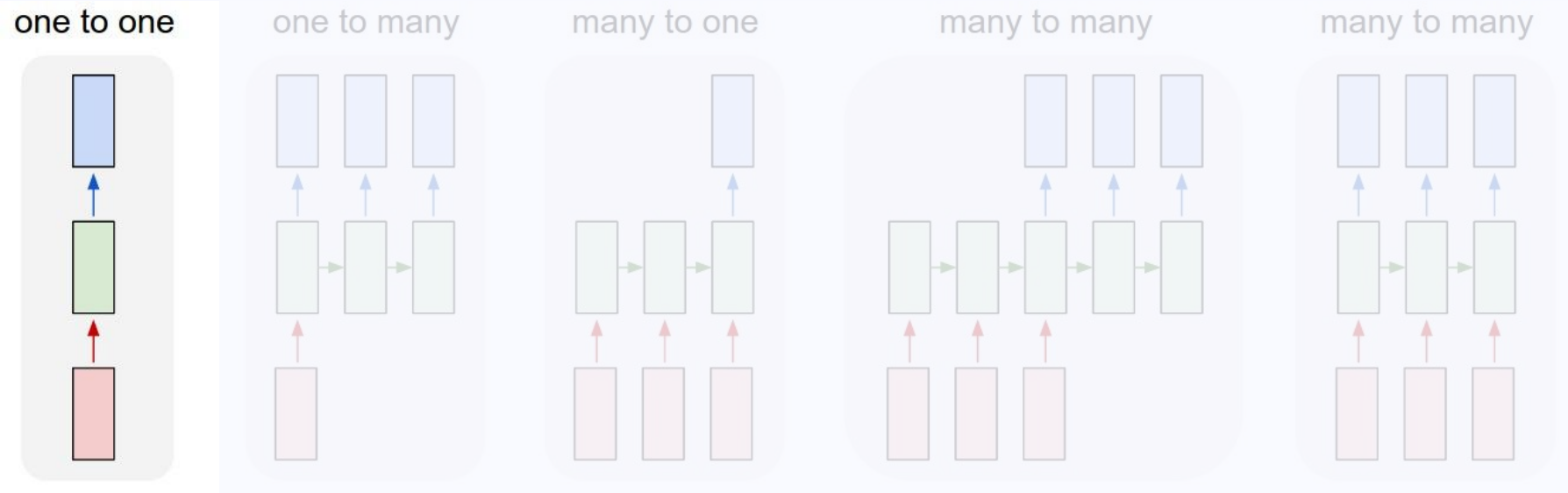
Convolutional Neural Networks

- **Feature maps** to detect features /patterns in the input data → NN
- (Some) drawbacks for NLP:
  - Fixed input size
  - Problem with long-distance relationships



https://domino.ai/blog/gpu-accelerated-convolutional-neural-networks-with-pytorch

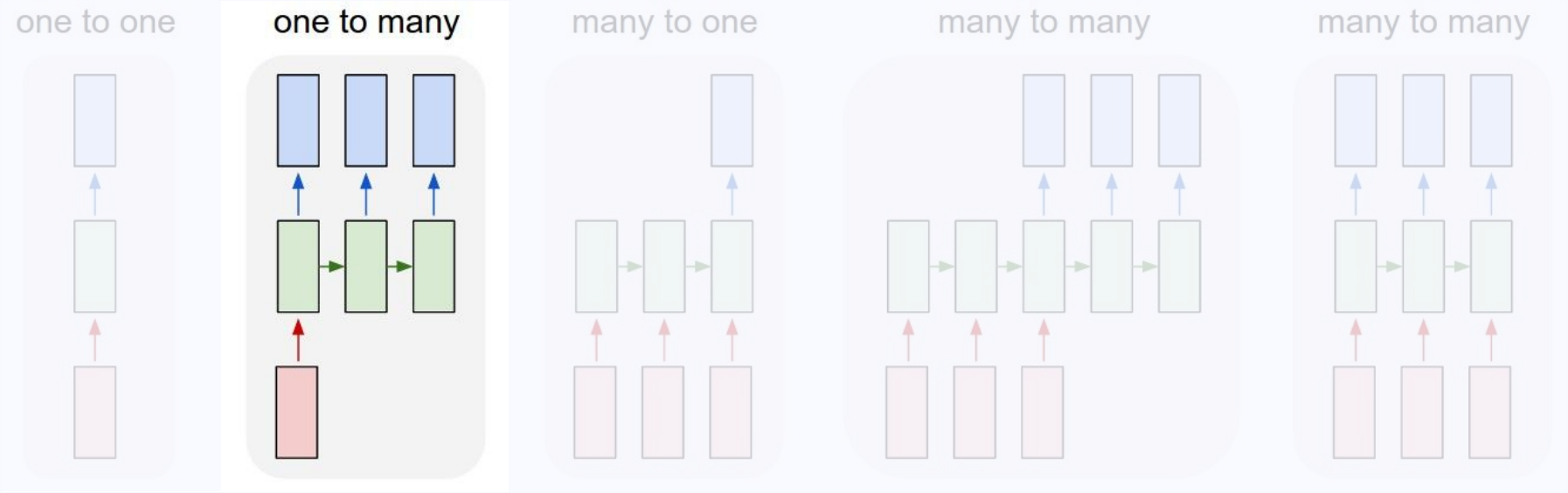
Structure of a NN



One fixed-size input → One fixed-size output (e.g. image classification)

https://karpathy.github.io/2015/05/21/rnn-effectiveness/

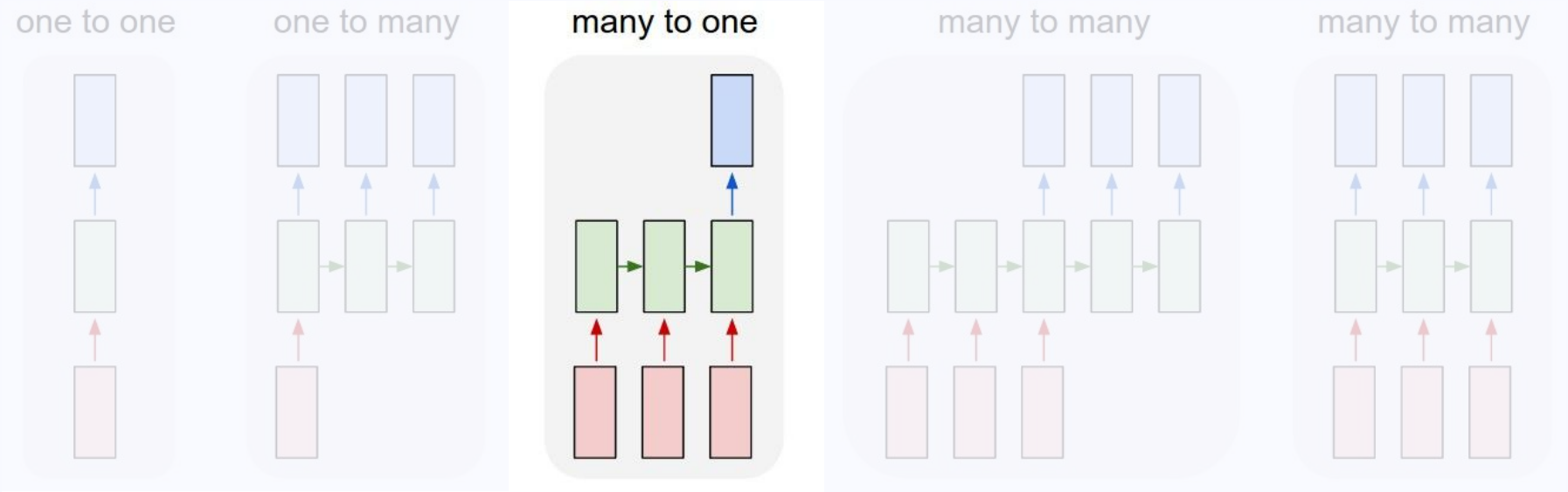
Structure of a NN



One fixed-size input → Output with multiple values (e.g. image captioning)

https://karpathy.github.io/2015/05/21/rnn-effectiveness/

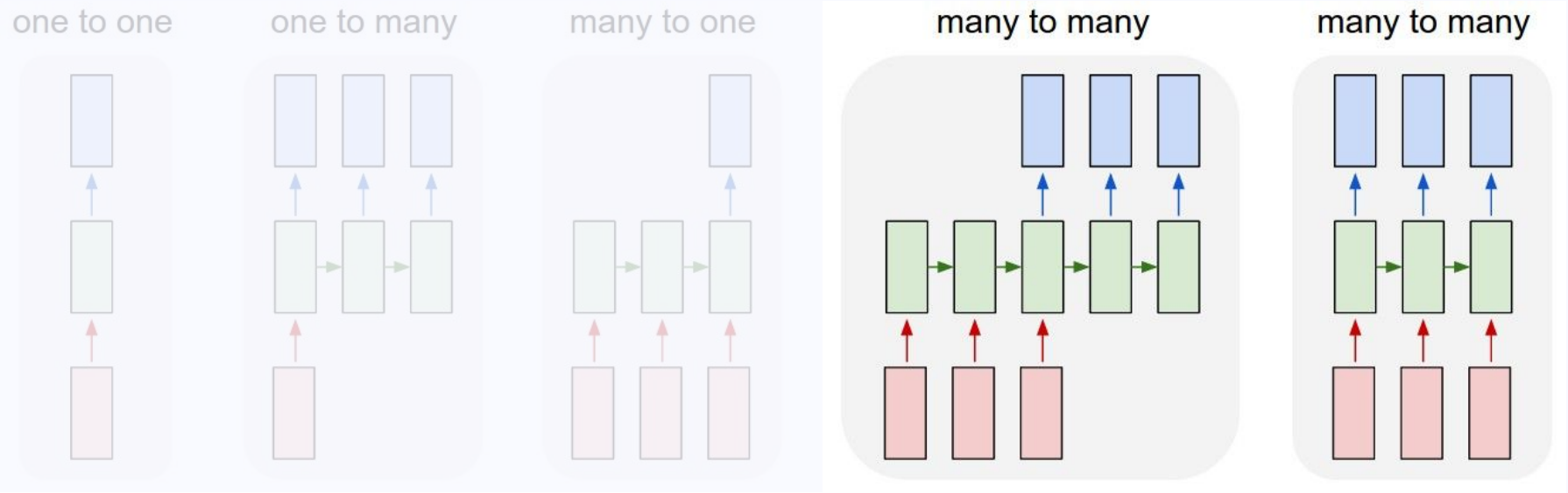
Structure of a NN



Input with multiple values → One fixed output (e.g. text classification)

https://karpathy.github.io/2015/05/21/rnn-effectiveness/

Structure of a NN



Input with multiple values → Output with multiple values (e.g. machine translation)

https://karpathy.github.io/2015/05/21/rnn-effectiveness/

2. Recurrent Neural Networks

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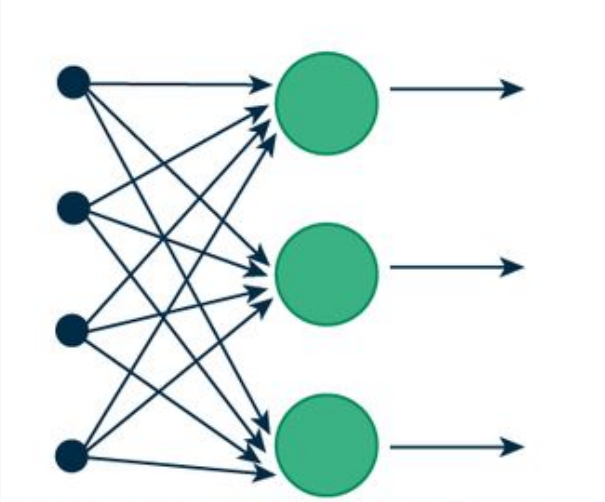
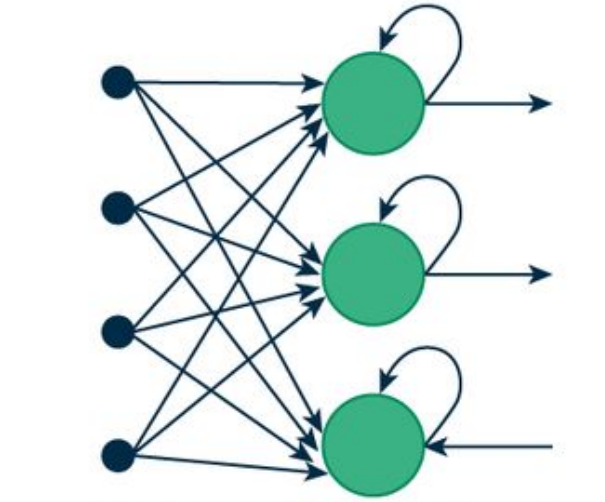
Introduction to RNNs

What is an RNN?

- Like a feedforward NN… but better with **feedback**!
- **Loops** that allow information to go back into the network

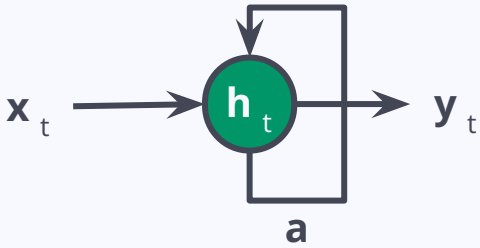


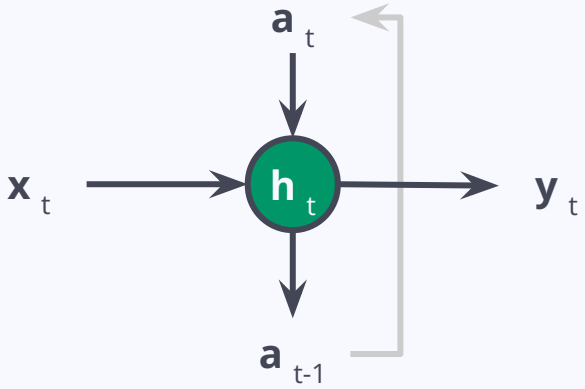


https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

Unfolding an RNN

- **Recurrent Unit**: Neuron that holds a **hidden state** that contains the past
  - $\mathbf{x}_t$ = **input** in *time t*
  - $\mathbf{y}_t$ = **output** in *time t*
  - $\mathbf{a}_{t-1}$ = **activation** in *time t-1* = the previous activation
  - $\mathbf{a}_t$ = **activation** in *time t* = the current activation
  - $\mathbf{h}_t$ = **hidden state** of the neuron in *time t*

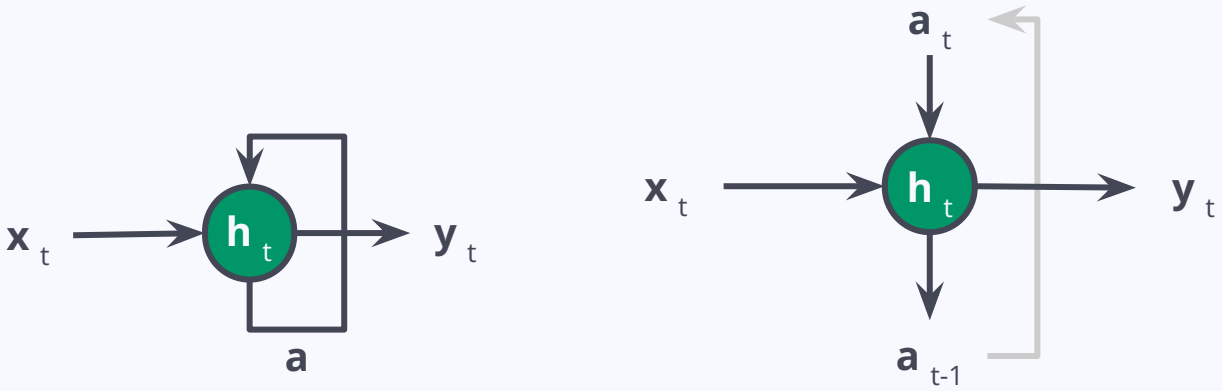




https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks

Unfolding an RNN

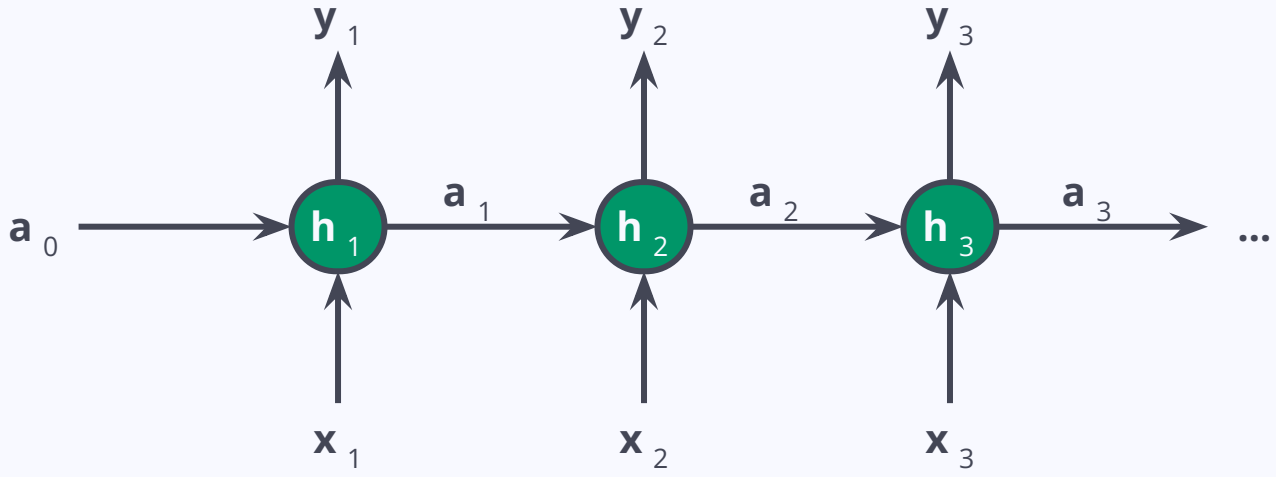
- **Backpropagation Through Time (BPTT)**: “learning from its errors”
- Current output = current output + memory of past inputs



https://www.geeksforgeeks.org/ml-back-propagation-through-time/

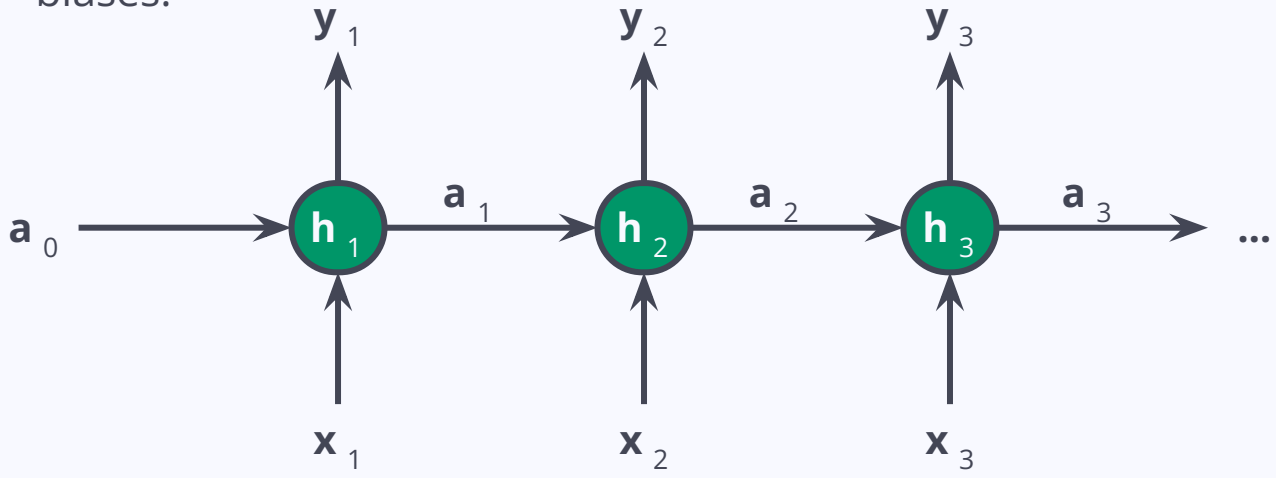
Structure of an RNN

- **Hidden state (h_t) calculation**: $h_t = \sigma(U \cdot x_t + W \cdot h_{t-1} + B)$
  - U, W = weights!
  - B = biases!



Structure of an RNN

- **Output (y_t) calculation**: $y_t = O\,(V \cdot h_t + C)$
  - O = activation function
  - V = weights!
  - C = biases!



RNNs: Applications

- Processing **sequential data**
  - Time-series:
  - NLP : words, sentences, etc. → Let’s focus on this…

RNNs & NLP

A Game of Sequential Data

- Recite the alphabet in your head...
  - In the right order!

A Game of Sequential Data

- Recite the alphabet in your head...
  - In the right order! → A, B, C, D, E, F, G, H, ...
  - Backwards!

A Game of Sequential Data

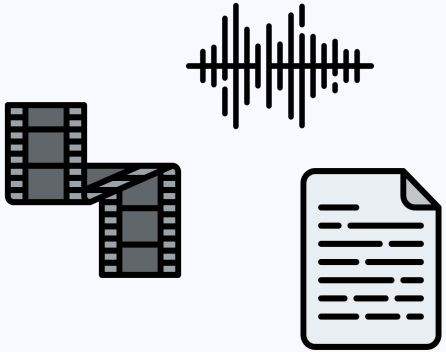
- Recite the alphabet in your head...
  - In the right order! → A, B, C, D, E, F, G, H, ...
  - Backwards! → Z, Y, X, W, V, U, ...
  - Starting from the letter “N” →

A Game of Sequential Data

- Recite the alphabet in your head...
  - In the right order! → A, B, C, D, E, F, G, H, ...
  - Backwards! → Z, Y, X, W, V, U, ...
  - Starting from the letter “N” → N, O, P, Q, R, S, T, ...

- We have learned the alphabet as a **sequence**
- Difficult in other ways!

Sequential data



- Video, audio, **text** are sequences
- But remember… 🍎🥧 ≠ 🍎📱

- We need a model that can:
  - handle context & position in the sentence
  - handle sentences with varying lengths
  - handle context left and right
  - handle long-distance relations

NLP with RNNs: Example

Binary Classification of a sentence: “What time is it?”

- Asking for the time? → Category 0
- Asking for the weather? → Category 1

The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

NLP with RNNs: Example

- Tokenization

The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

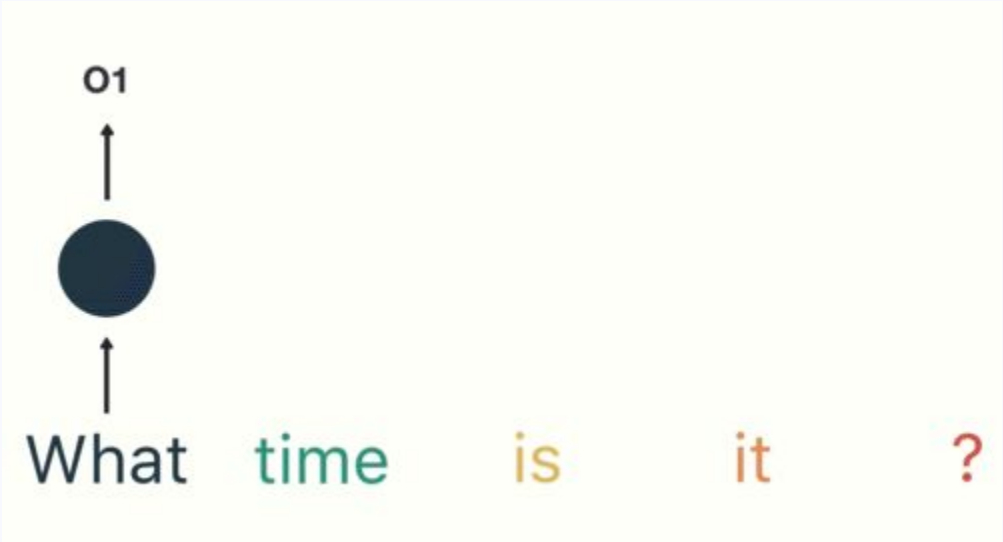
NLP with RNNs: Example

- **1st timestep**: pass the word “What” to the model

The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

NLP with RNNs: Example

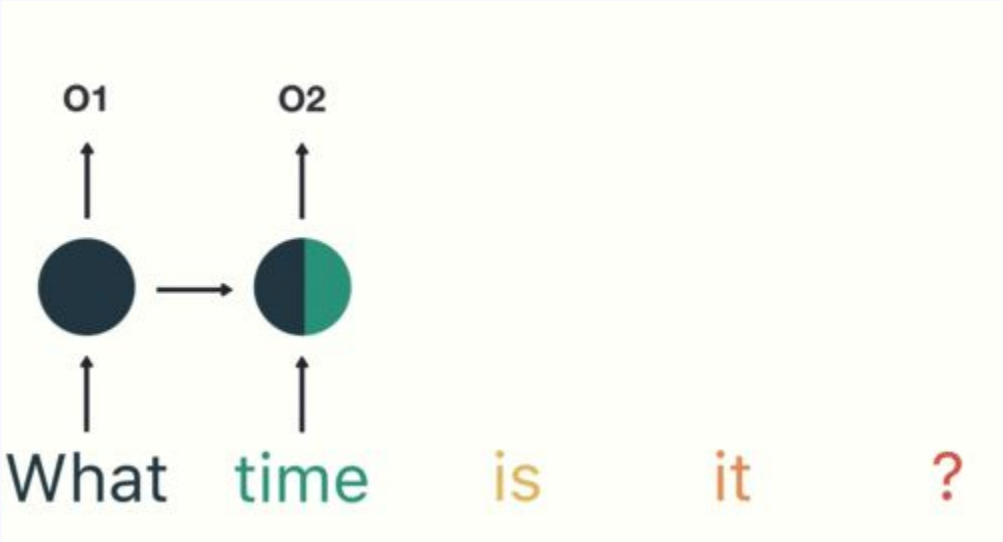
- **2nd timestep**: pass the word “time” and the hidden state with “what”



The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

NLP with RNNs: Example

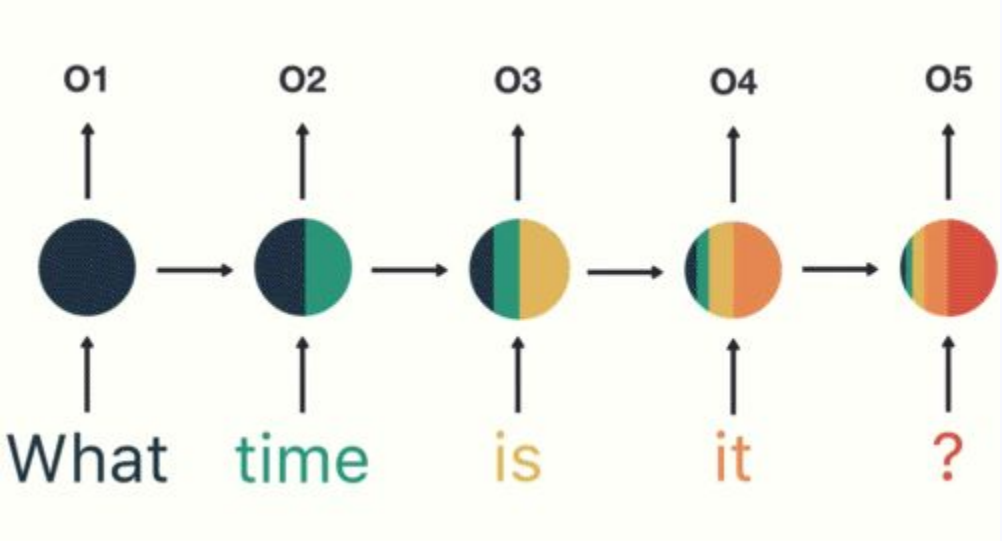
- Repeat for all timesteps until end of sequence



The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

NLP with RNNs: Example

- Binary Classification: O_5 = **Probabilities** of Category 0 and Category 1



The AI Hacker: https://youtu.be/LHXXI4-IEns?si=wHLRrAlpqLUFRzGP

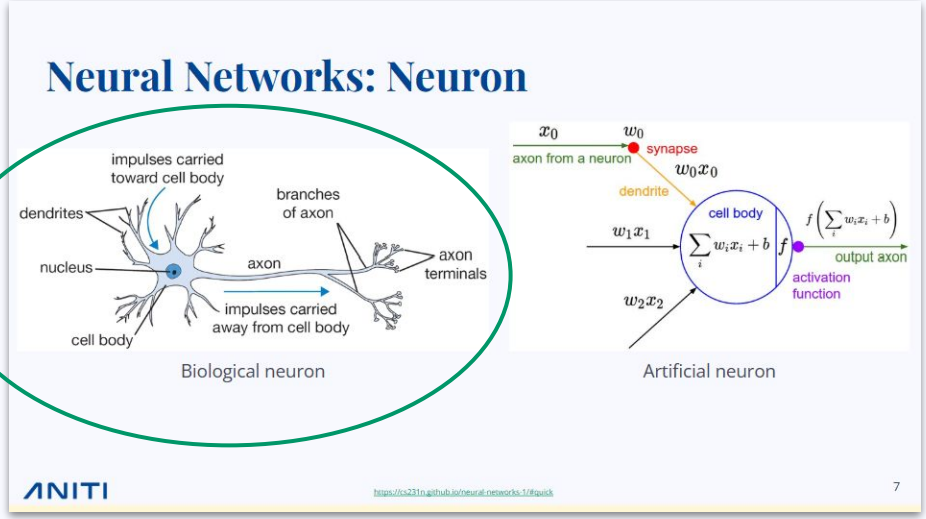
Vanishing gradient problem

Do you remember what was in the first image in Slide 7?

Vanishing gradient problem

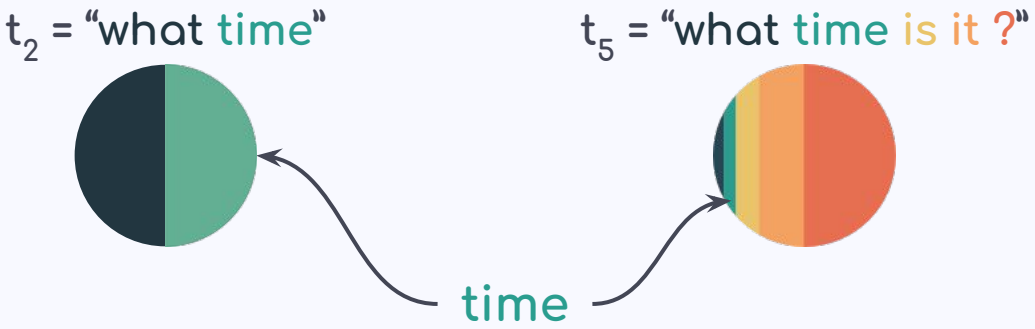
Do you remember what was in the first image in Slide 7?

Human working memory has limits… and so do RNNs



Vanishing gradient problem

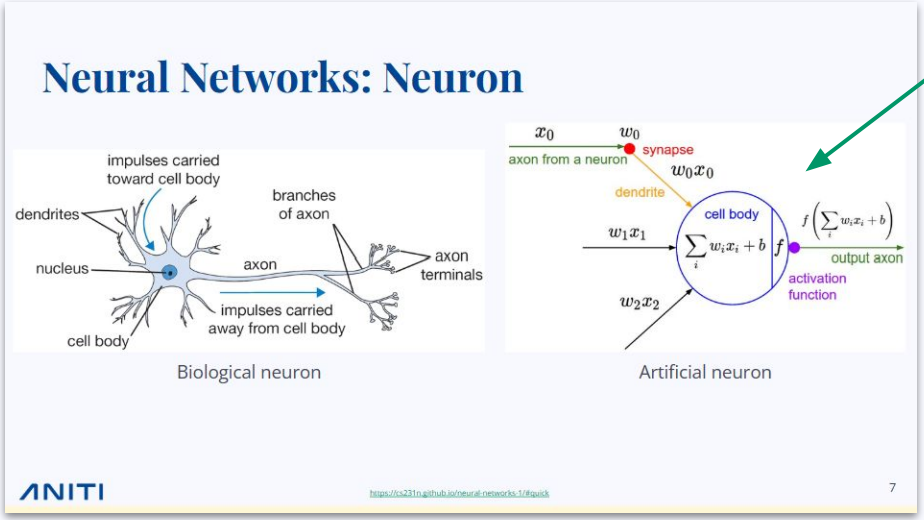
- **BPTT**: gradients exponentially shrink as timesteps pass
- Final timestep: early hidden states are “not important” anymore



Exploding gradient problem

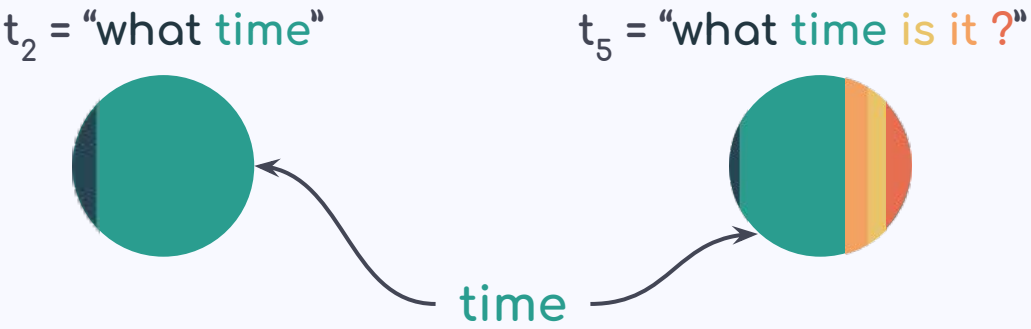
Now you remember what was in the first image in Slide 7!

...but it's not so important to our lecture today!



Exploding gradient problem

- What if we change the weights to make some hidden states important…?
- “time” is now very important… but this could be a problem with more data!



RNNs: Pros & Cons

+ Processing input of any length

− Computation being slow

https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks

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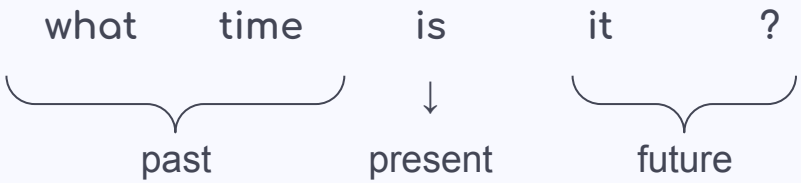
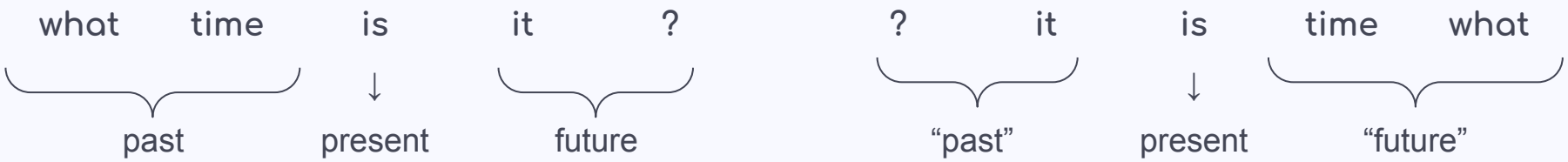
Bidirectionality

- NNs process input left → right: past → present → ~~future~~
- But what if we **reversed the sequence**?

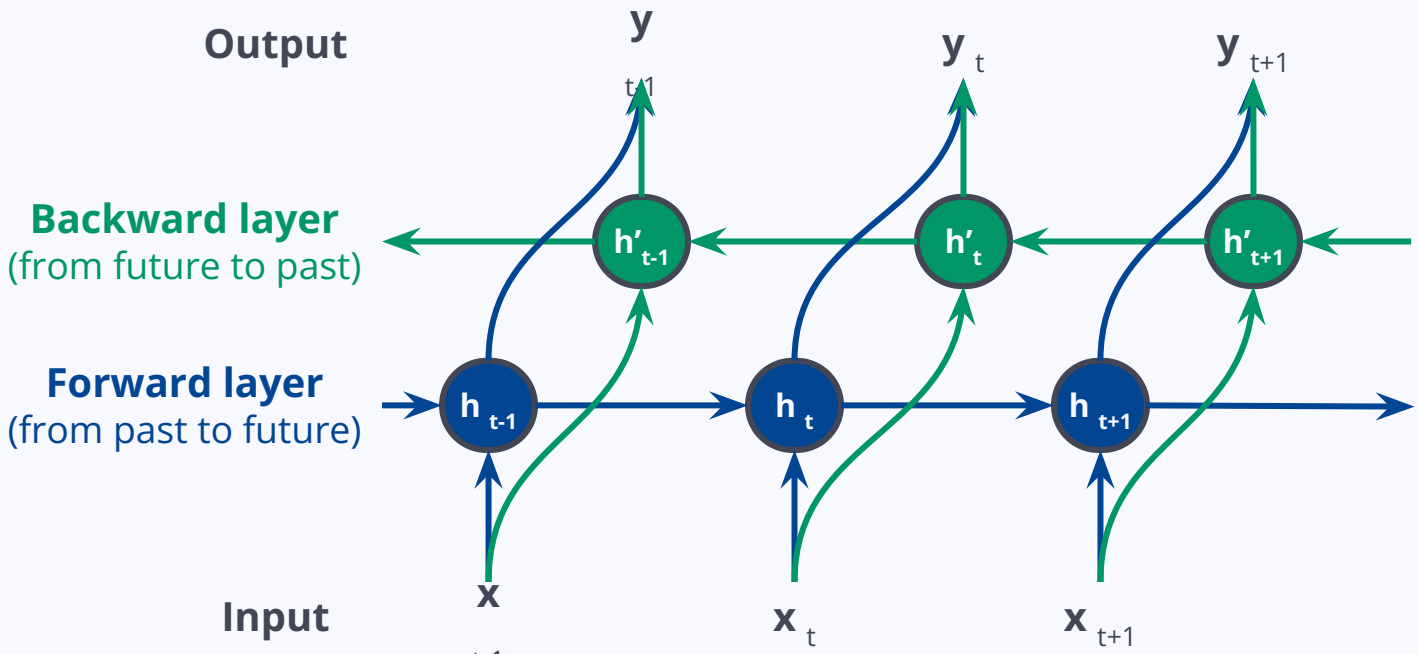


Bidirectionality

- NNs process input left → right: past → present → ~~future~~
- But what if we **reversed the sequence**?
- The future is to the left → accessible to the model as a “past” state!

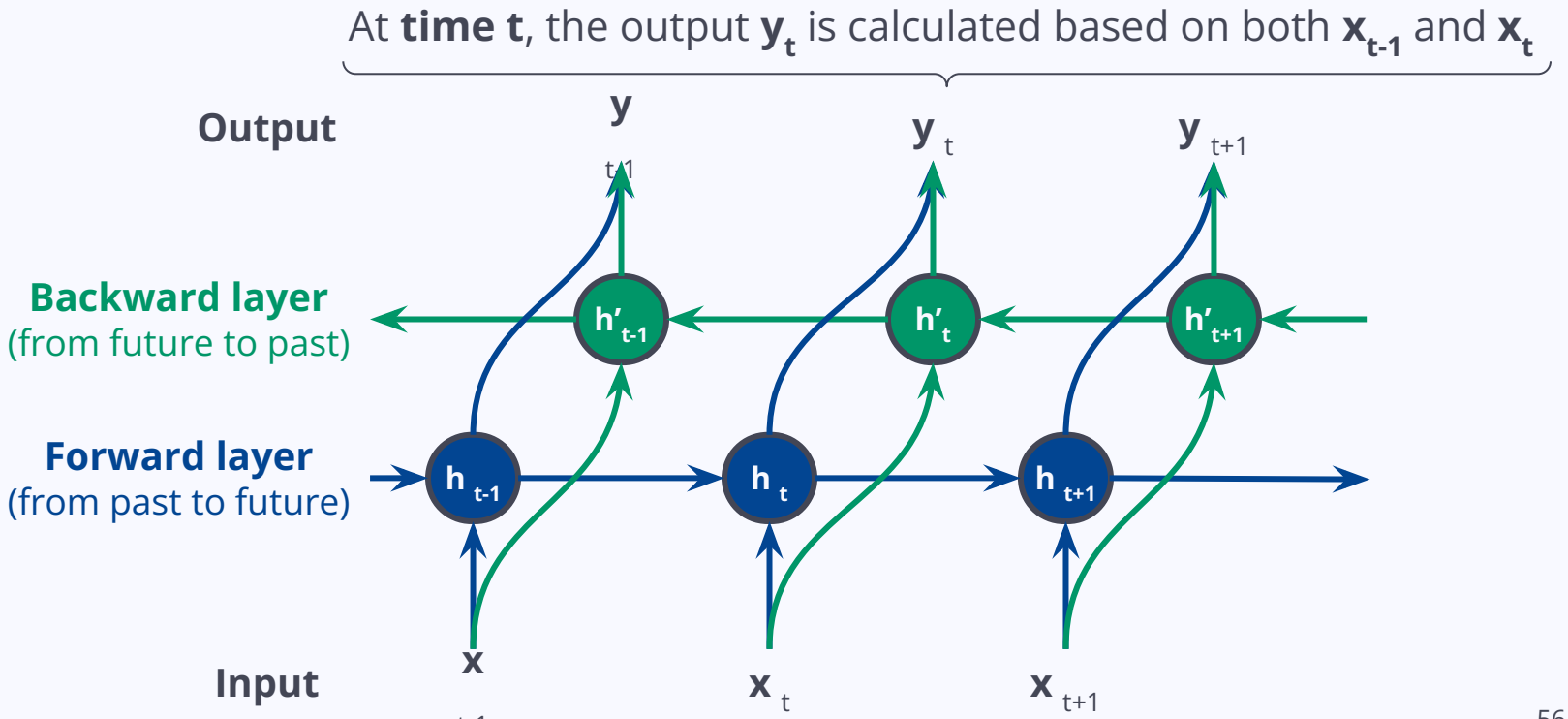


Bidirectionality: bi-RNN



https://www.geeksforgeeks.org/bidirectional-recurrent-neural-network/

Bidirectionality: bi-RNN



https://www.geeksforgeeks.org/bidirectional-recurrent-neural-network/

RNNs: Pros & Cons

+ Processing input of any length
+ Modeling using **past** samples, capture dependencies!
+ Weights are shared across time

– Computation being slow
– Cannot use any **future** input!

Improve the RNN architecture!

– Vanishing/Exploding gradients: difficulty with **long sequences**

https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks

3. Long Short-Term Memory Neural Networks

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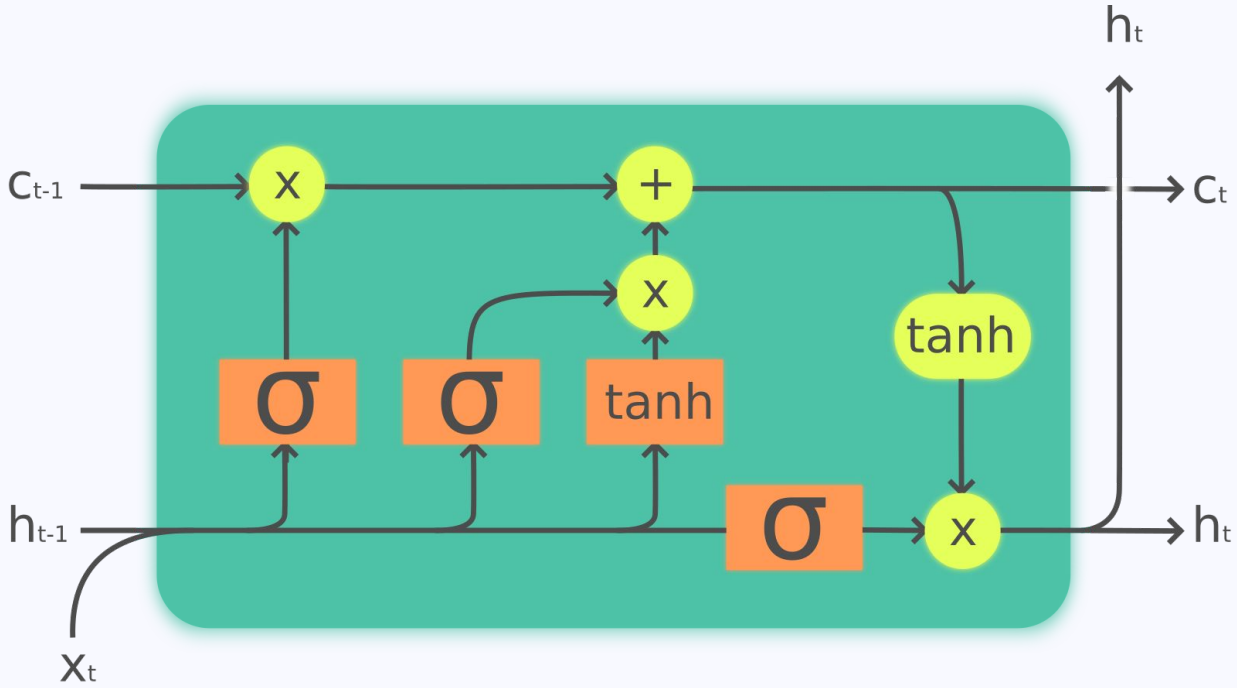
What is an LSTM?

- **Long Short-Term Memory** Neural Network: improvement on RNNs
- Designed to avoid vanishing/exploding gradients

- RNNs: one linear path for all previous states
- LSTMs:
  - one path for **long-term "memories"** (states) → directly influence output
  - one path for **short-term "memories"** (states) → follow the flow of information

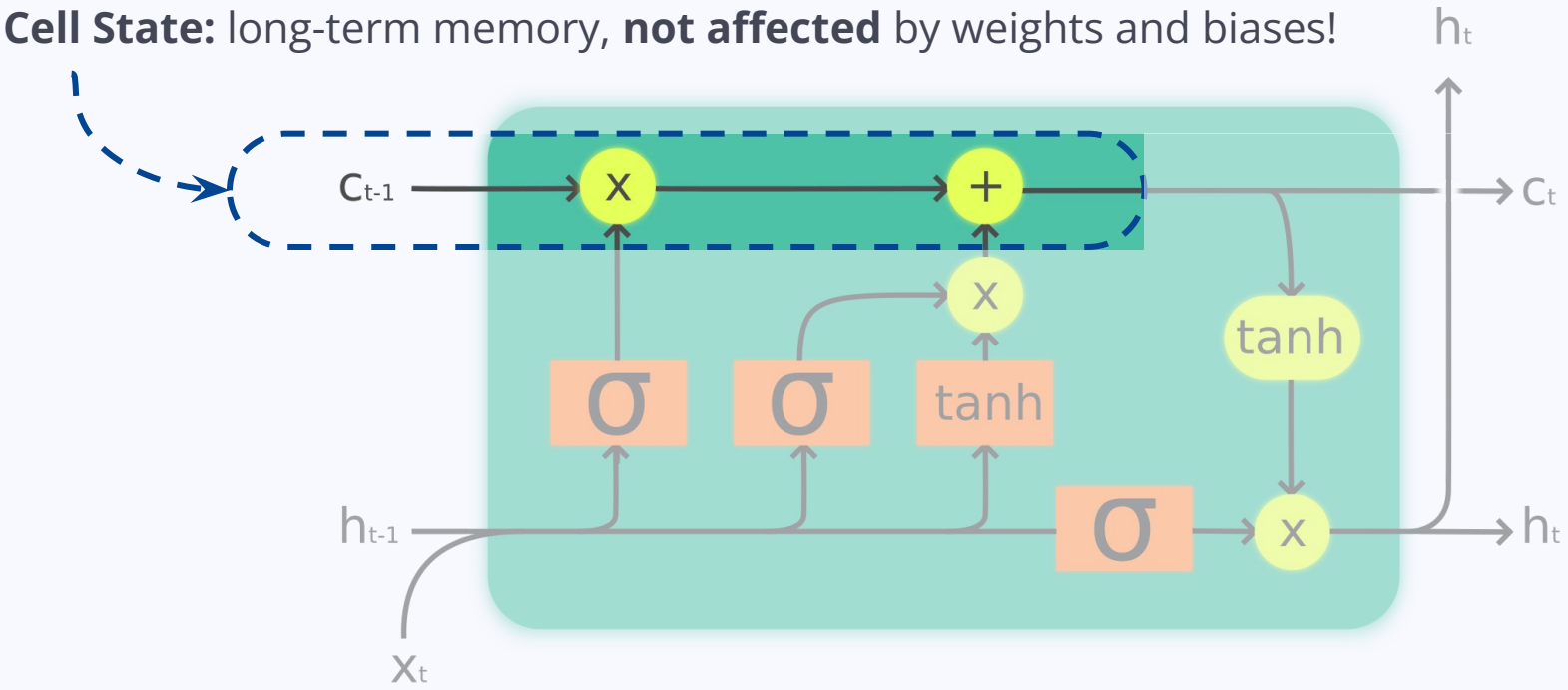
LSTM: Cell



https://commons.wikimedia.org/wiki/File:The_LSTM_cell.png

LSTM: Cell

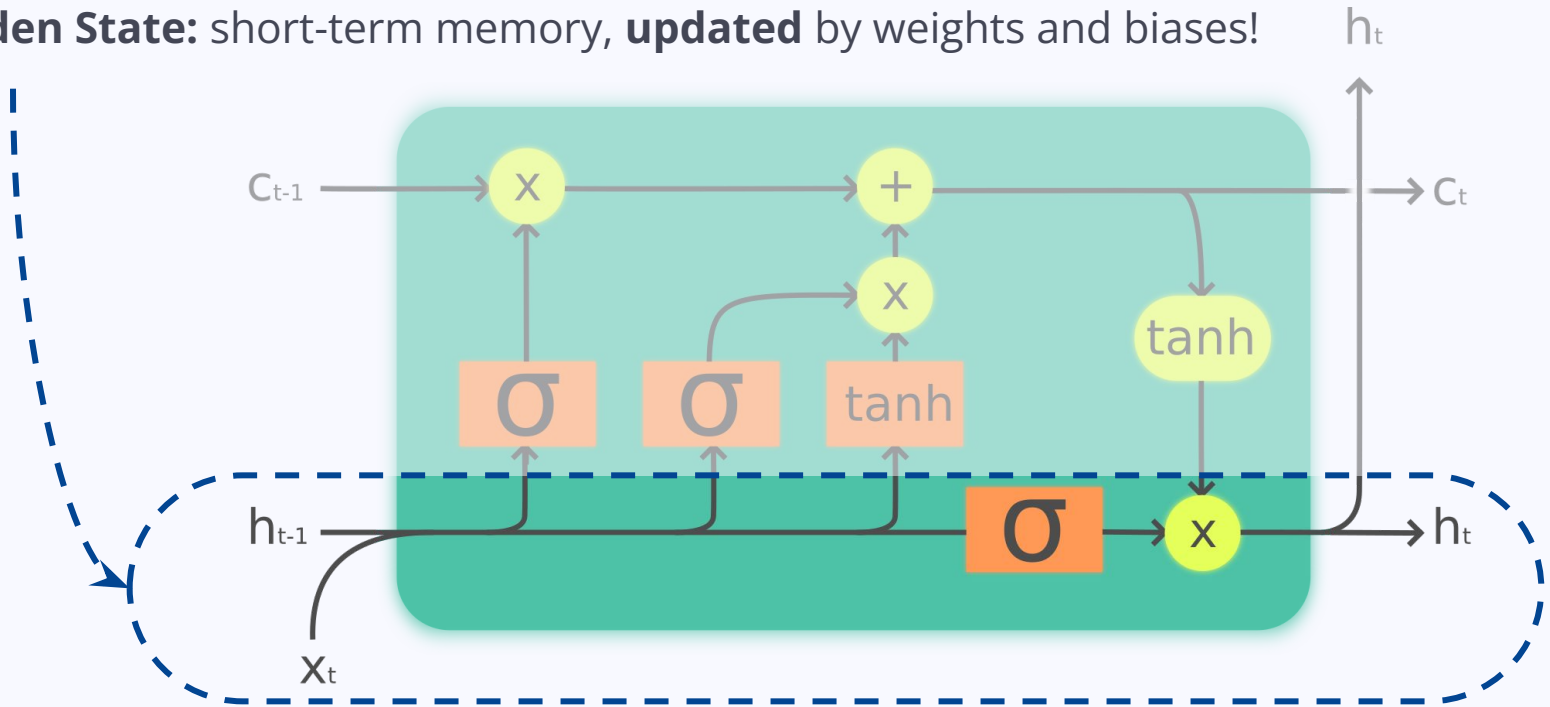
Cell State: long-term memory, **not affected** by weights and biases!



https://commons.wikimedia.org/wiki/File:The_LSTM_cell.png
StatQuest with Josh Starmer: https://youtu.be/YCzL96nL7j0?si=w_lyt6vzSxhBt9n9

LSTM: Cell

Hidden State: short-term memory, **updated** by weights and biases!



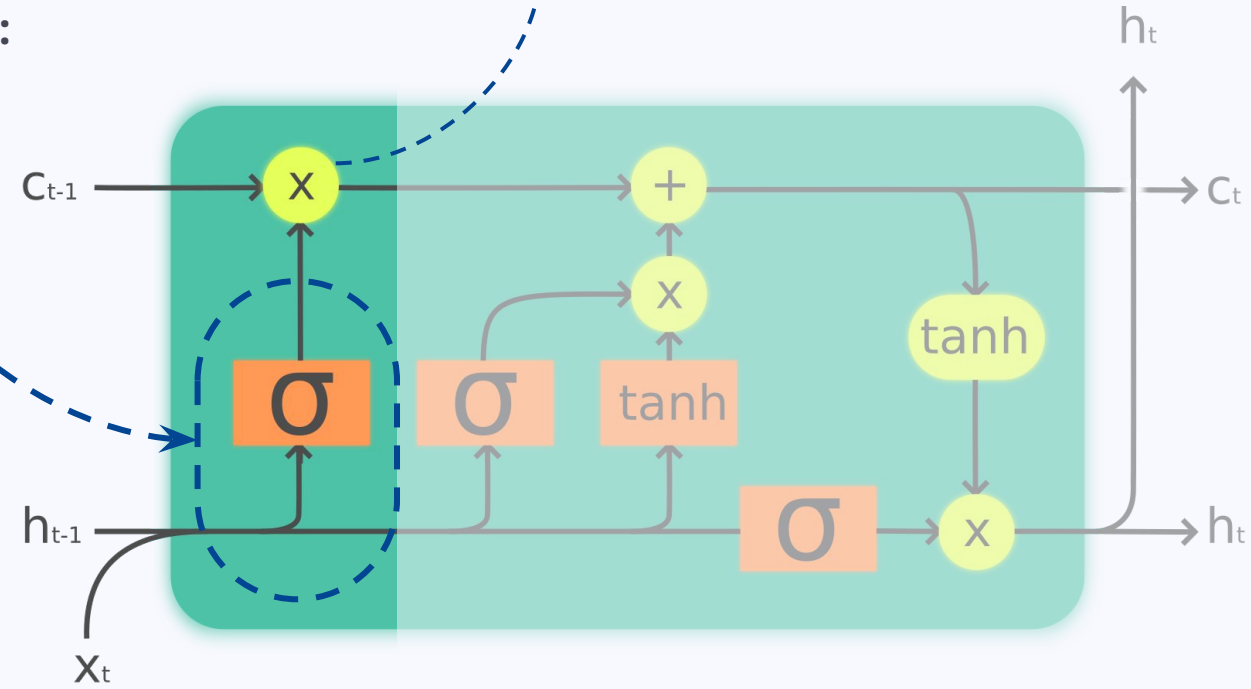
https://commons.wikimedia.org/wiki/File:The_LSTM_cell.png
StatQuest with Josh Starmer: https://youtu.be/YCzL96nL7j0?si=w_lyt6vzSxhBt9n9

LSTM: Cell

σ = sigmoid activation function (0-1)

x = % how much long-term memory to keep

Forget Gate:



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StatQuest with Josh Starmer: https://youtu.be/YCzL96nL7j0?si=w_lyt6vzSxhBt9n9

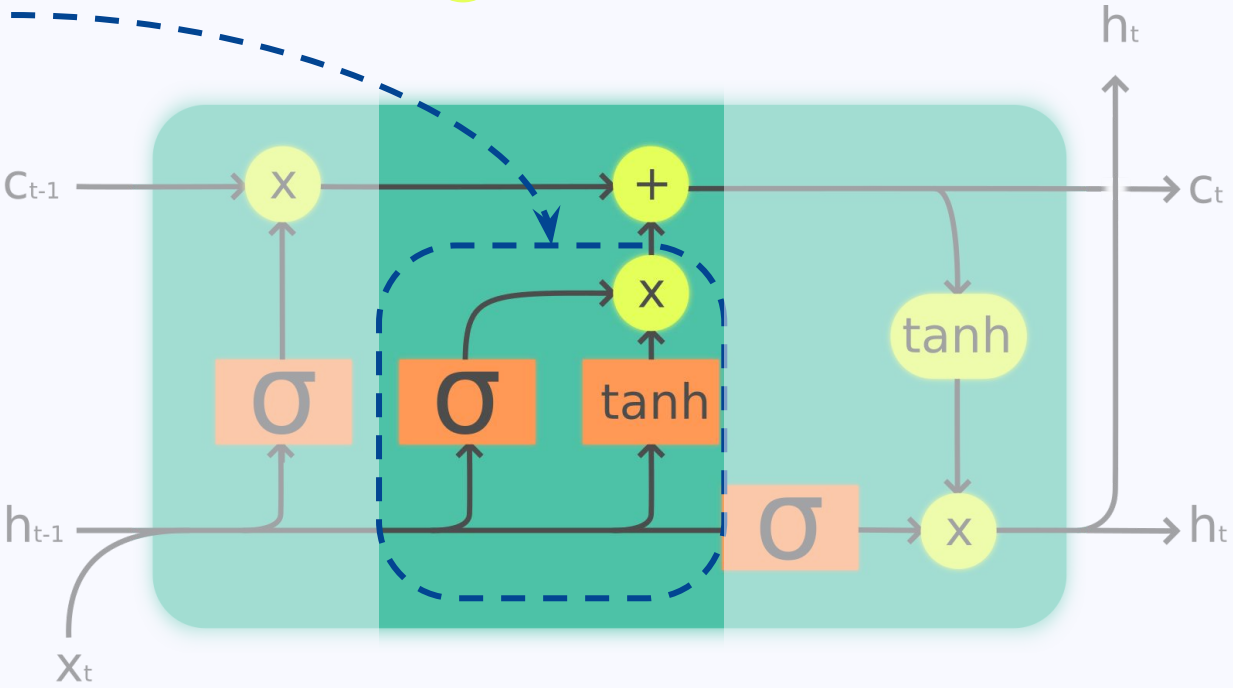
LSTM: Cell

tanh = tanh activation function (-1-1)

x = % of **new memory** to <u>update long-term memory</u>

\+ = add to the existing **long-term memory**

Input Gate:

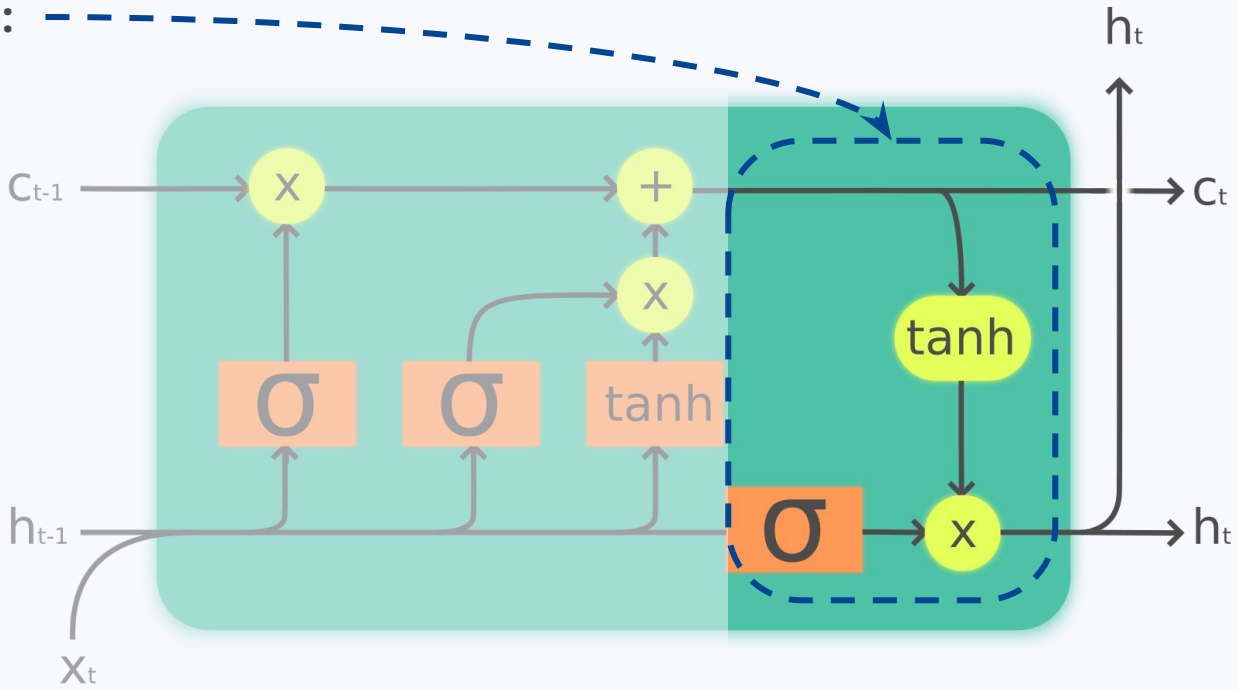


tanh = tanh with **new long-term memory**

x = % of **new memory** to <u>update short-term memory</u>

LSTM: Cell

Output Gate:



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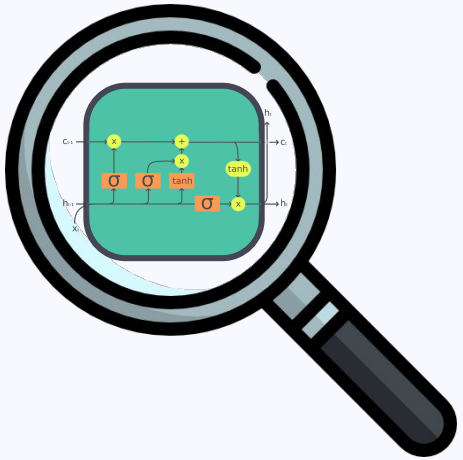
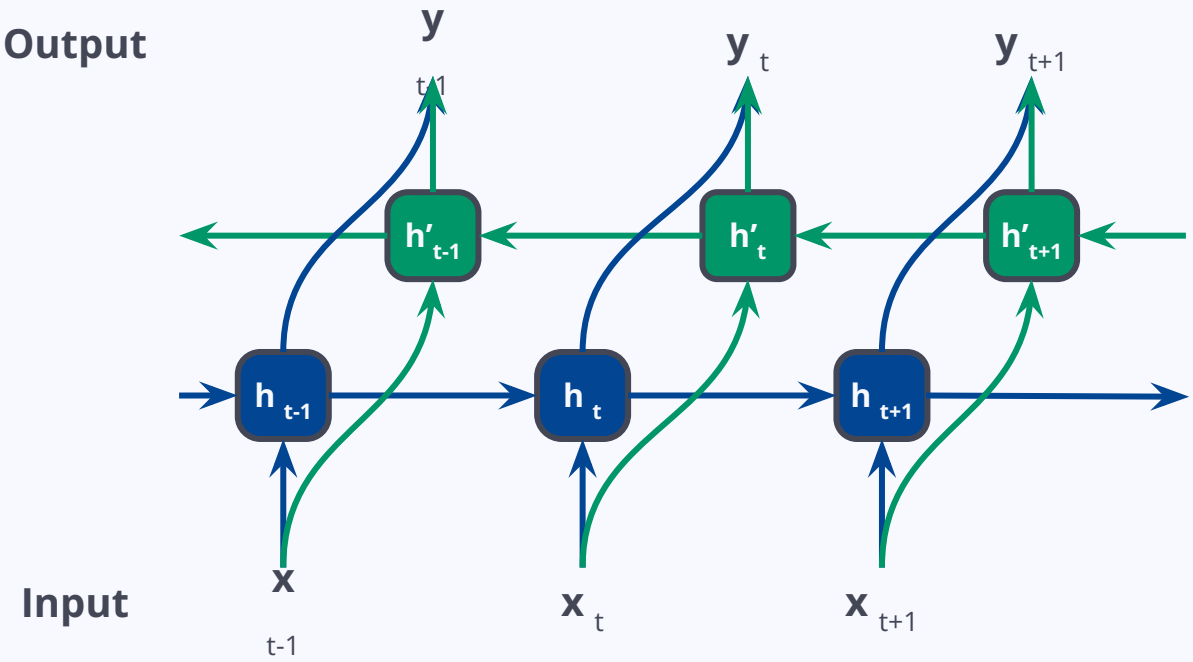
StatQuest with Josh Starmer: https://youtu.be/YCzL96nL7j0?si=w_lyt6vzSxhBt9n9

LSTM: Model

e.g. a bidirectional LSTM!





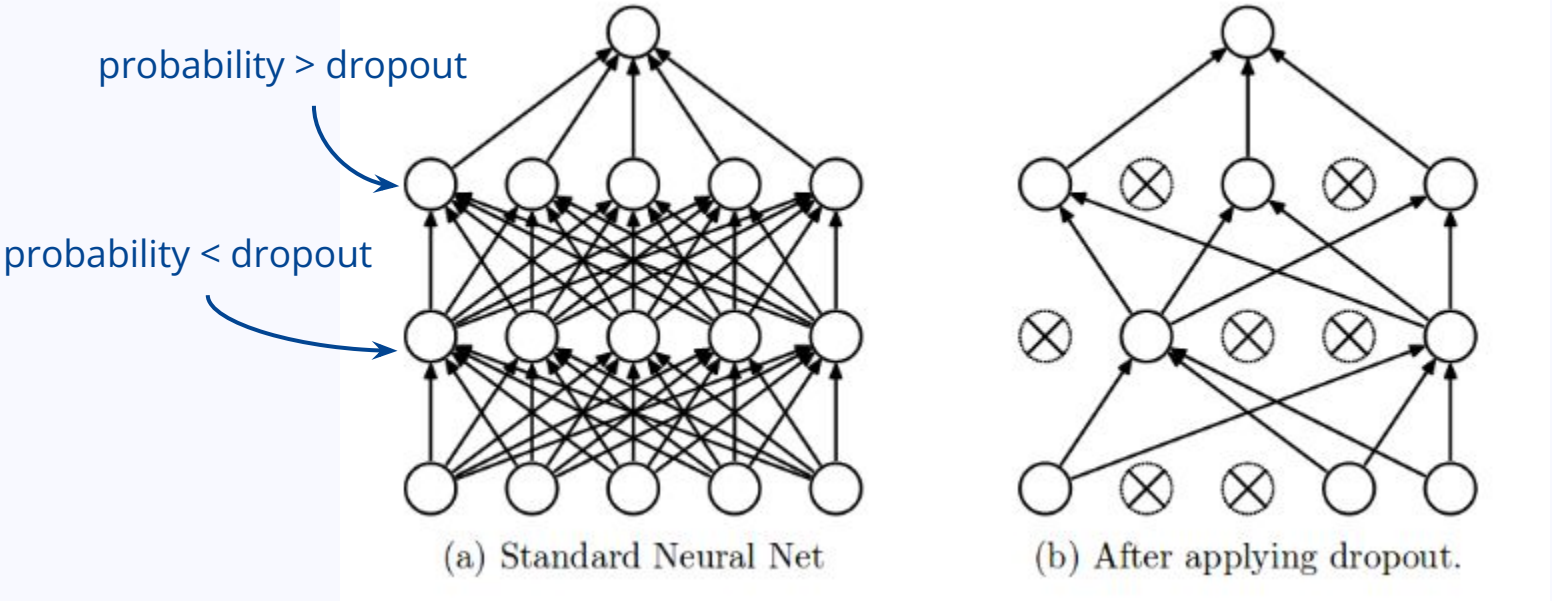
LSTMs: Pros & Cons

+ **Memory unit** to learn long-term dependencies in sequential data
+ **Robust** to noisy data
+ **Flexible** for many tasks

– Slower to train than RNNs because of extra calculations
– Vanishing/exploding gradients can still be a problem…
– Prone to **overfitting**

Dropout rate

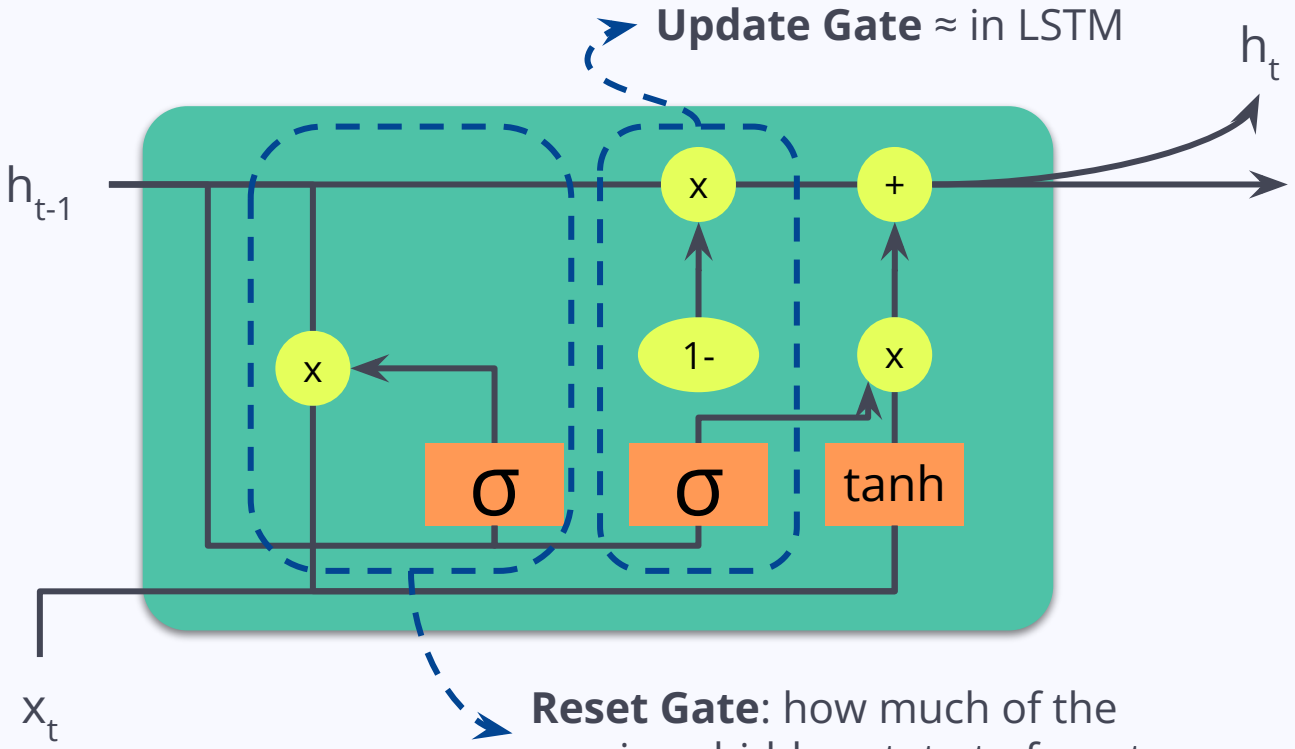
Temporarily “deactivate” some nodes of the input and hidden layer(s)



Srivastava et al. (2014): https://jmlr.org/papers/volume15/srivastava14a/srivastava14a.pdf

Gated Recurrent Units (GRU)

1- = inversion operation



4. Encoder-Decoder architecture

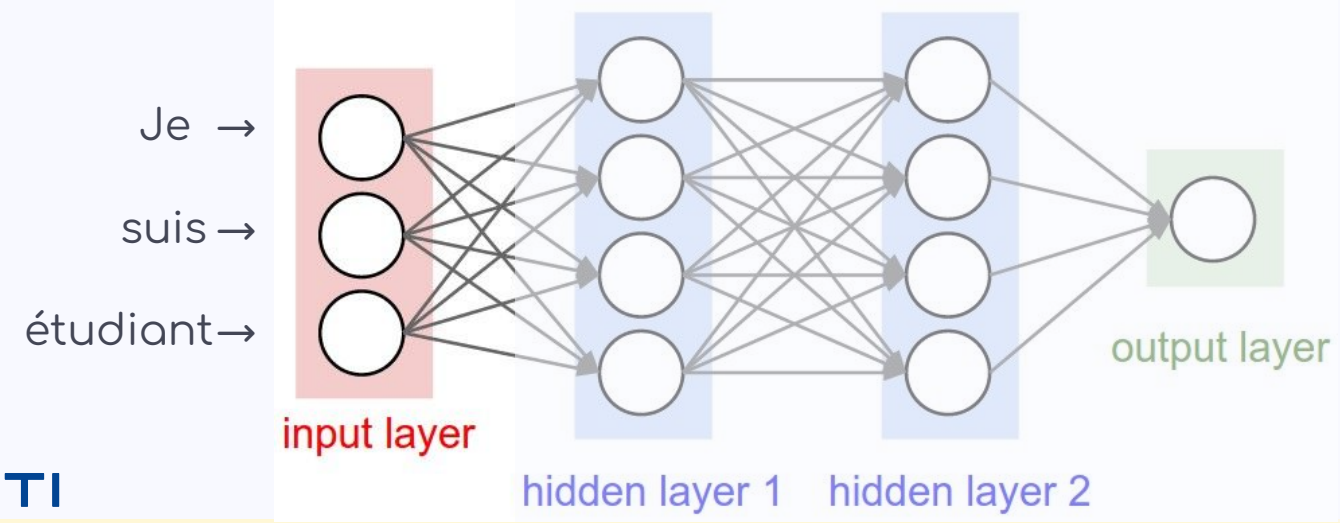
What is an encoder?

- **Part** of a neural network architecture
- Made of feedforward NNs, RNNs, LSTMs, or GRUs, etc...
- Turns a sequence into numerical representations = **context vectors**
- Pure encoder model? Yes, e.g. the sentence classifier!

https://library.fiveable.me/natural-language-processing/unit-8/encoder-decoder-architecture/study-guide/0Ku3zt66xYRvLfvu

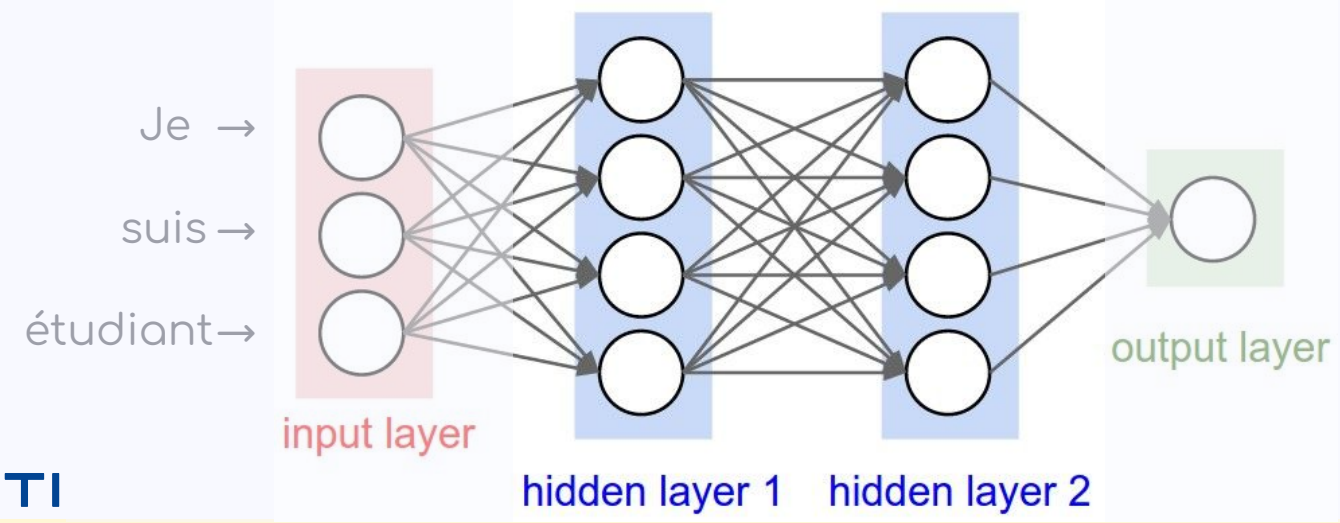
What is an encoder?

1. **Encode** each input token



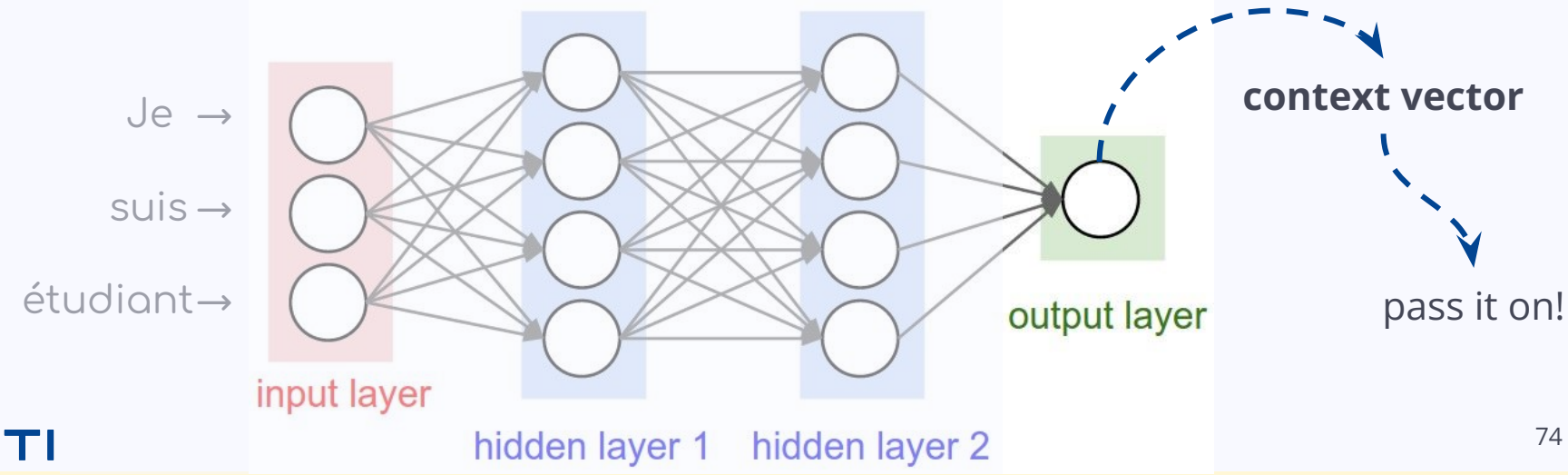
What is an encoder?

1. **Encode** each input token
2. 🕐→ 🕐 → 🕐: **update** hidden state based on input + context (same w&b!)



What is an encoder?

1. **Encode** each input token
2. 🕐→ 🕐 → 🕐: **update** hidden state based on input + context (same w&b!)
3. Encoder output = its final hidden state = **context vector**



Encoder's context vector

- Remember word embeddings? e.g. word2vec
- **Representation of the input sequence**
- Reminder: we use these context vectors as a plug-in for other models!
- Spoiler: we can make some very powerful word embeddings with this information about a word's context…

https://library.fiveable.me/natural-language-processing/unit-8/encoder-decoder-architectur…ge/0Ku3zt66xYRvLfv

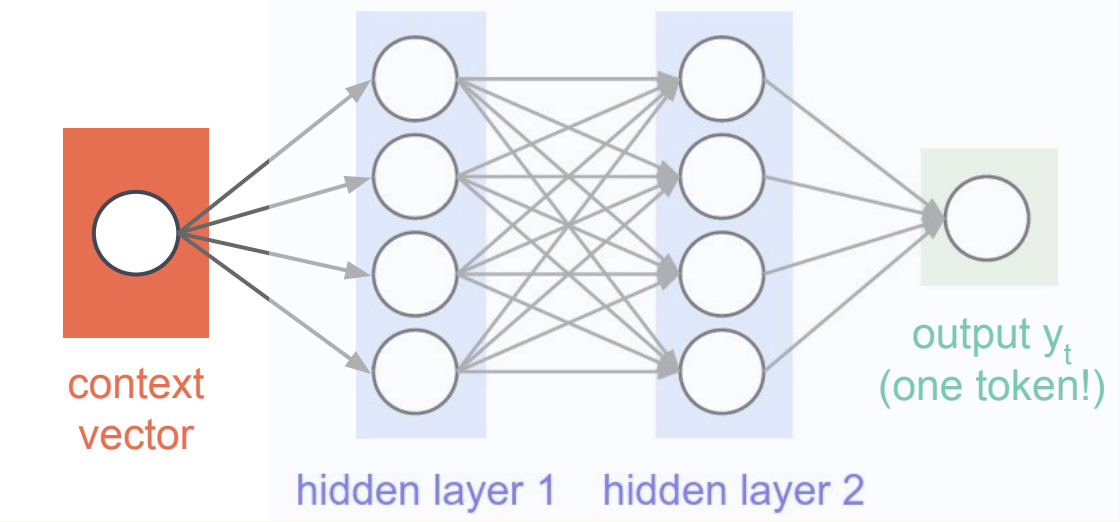
What is a decoder?

- **Part** of a neural network architecture
- **Reconstruct the encoded representation**
- Work **one token** at a time by picking the token with the highest probability
- **Autoregressive generation**: predicting a token based on the previously generated tokens
- Pure decoder model? Yes!

https://library.fiveable.me/natural-language-processing/unit-8/encoder-decoder-architecture/study-guide/0Ku3zt66xYRvLfvu

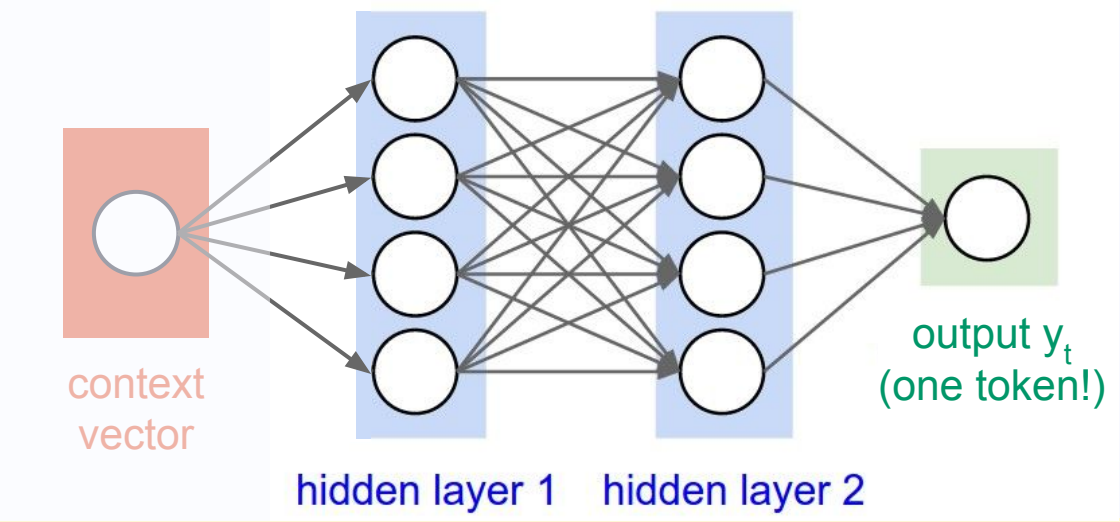
What is a decoder?

1. **Input**: the context vector as initial hidden state



What is a decoder?

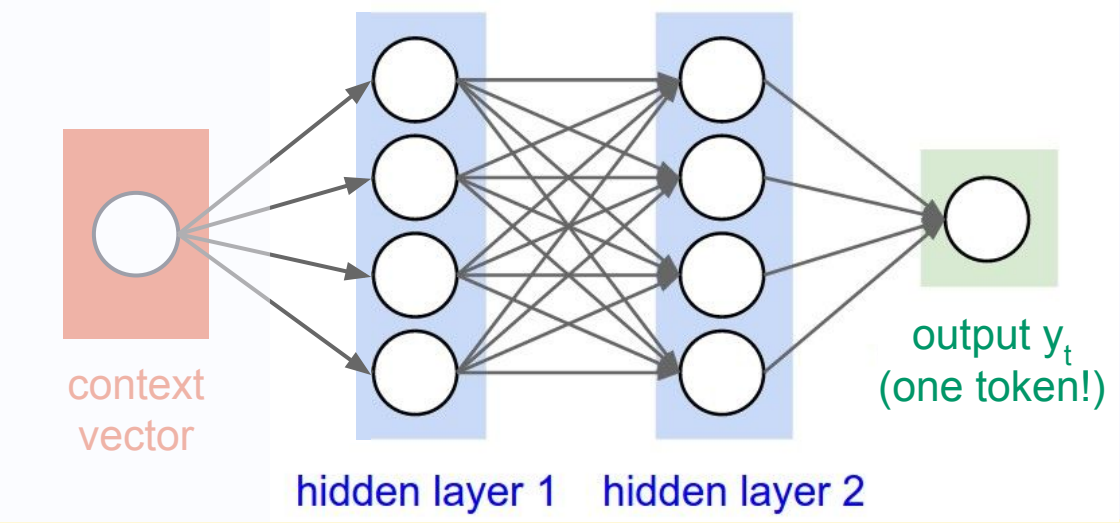
1. **Input**: the context vector as initial hidden state
2. 🕐→ 🕐 → 🕐 : **predict** next token based on current hidden state + previously generated tokens



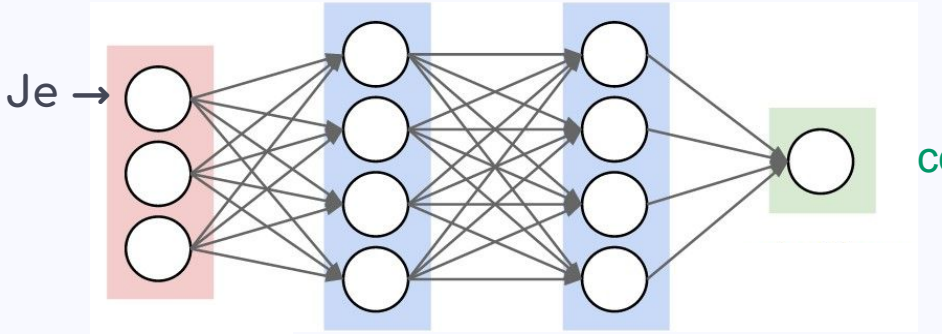
What is a decoder?

1. **Input**: the context vector as initial hidden state
2. 🕐→ 🕐 → 🕐 : **predict** next token
3. **Stop** when it predicts <eos> or max length reached!

End
of
Sequence



Encoder-decoder architecture



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[Machine Translation]

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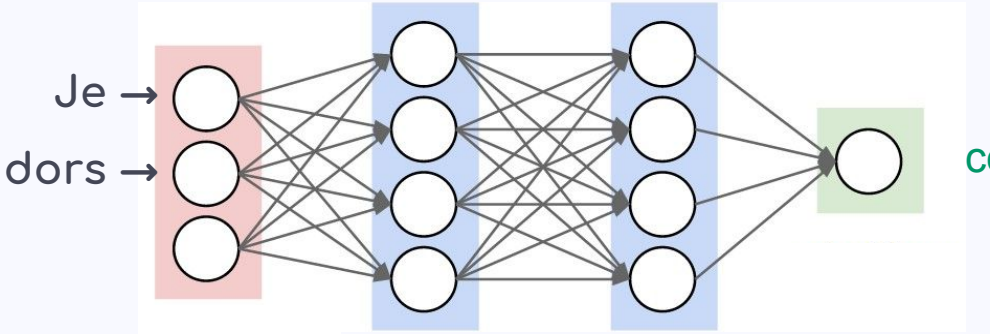
I am sleeping

Encoder-decoder architecture



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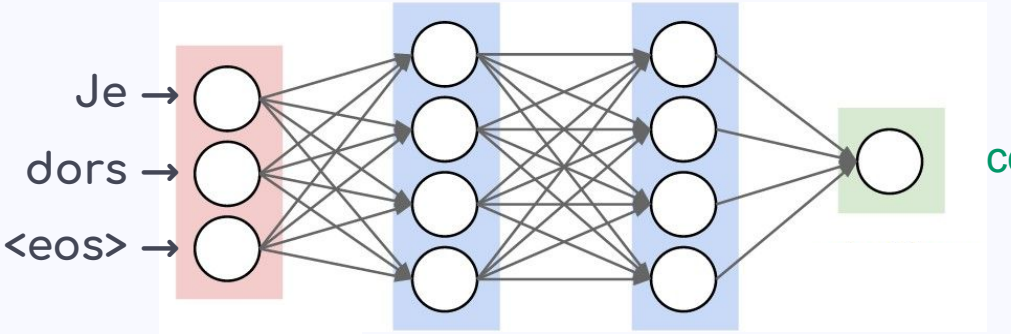

context vector

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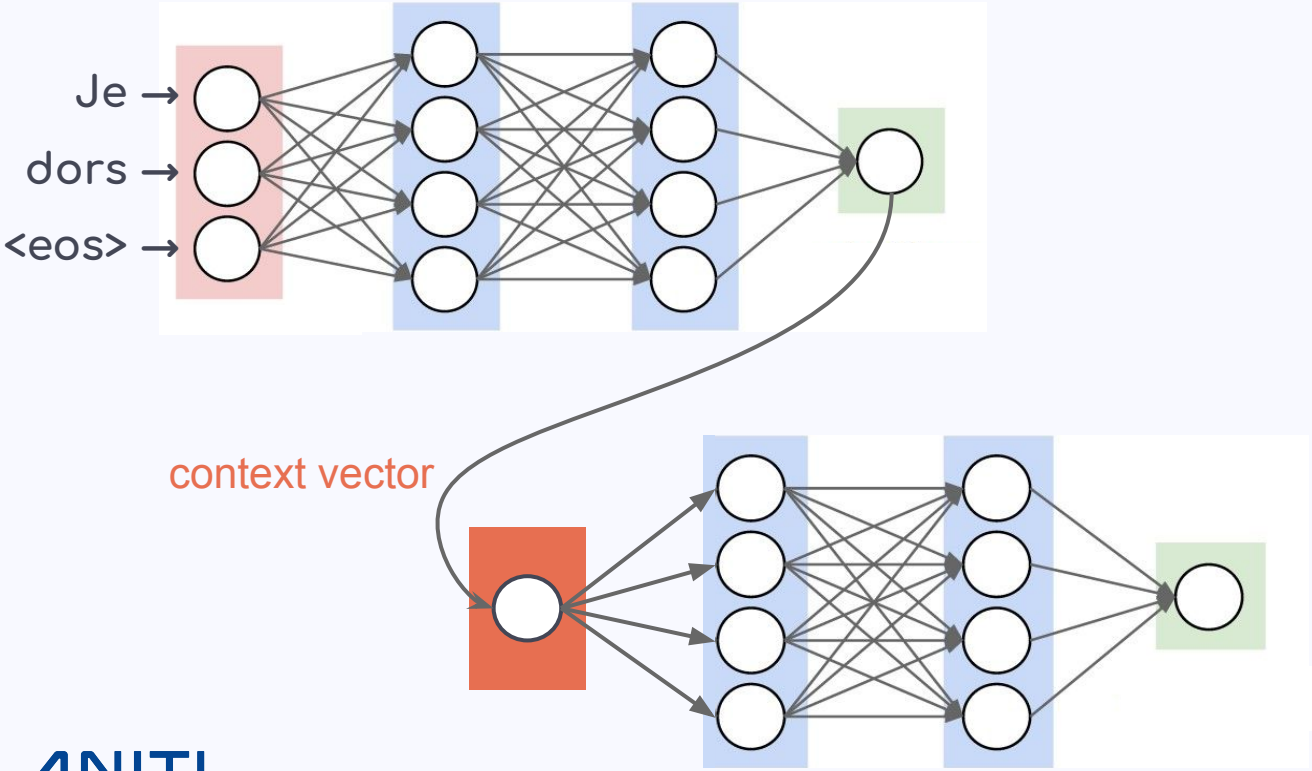
Encoder-decoder architecture



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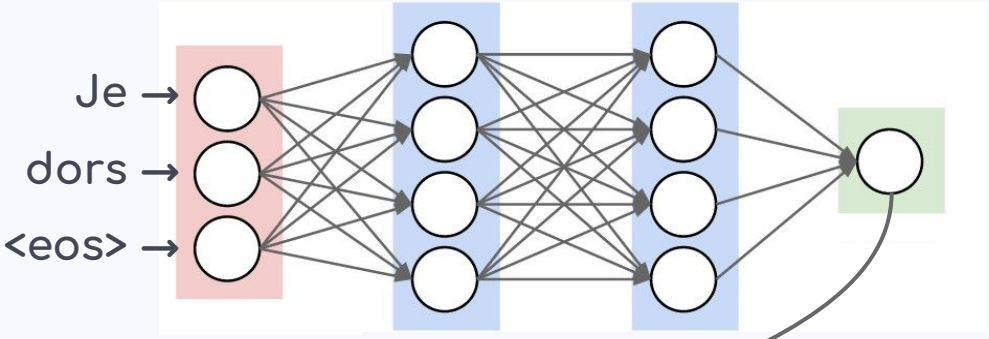
Encoder-decoder architecture



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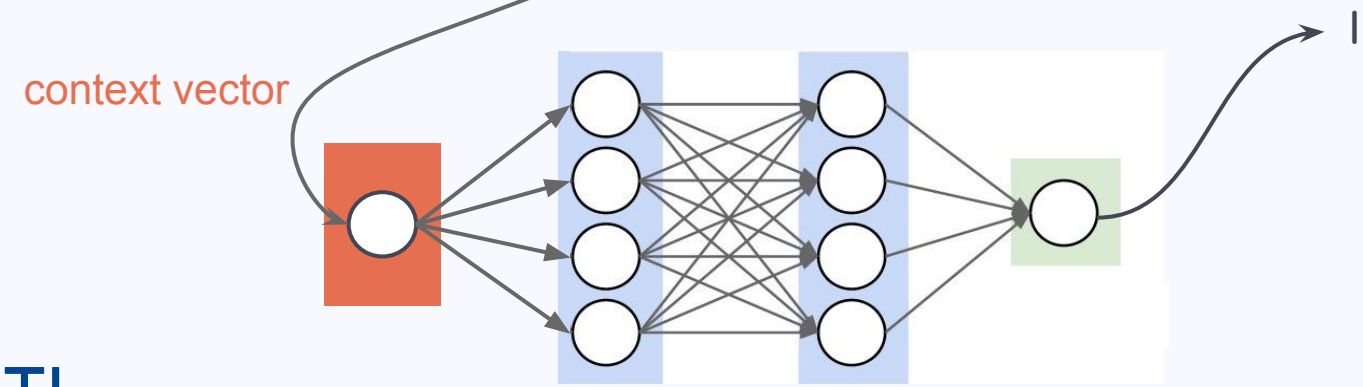
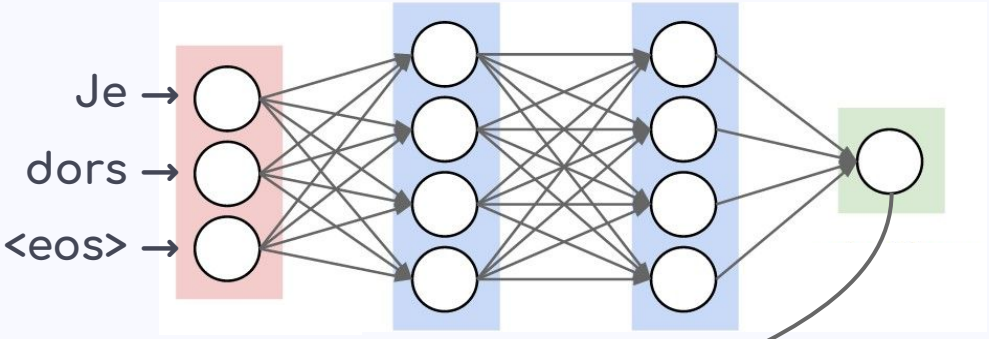


Encoder-decoder architecture



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[Machine Translation]

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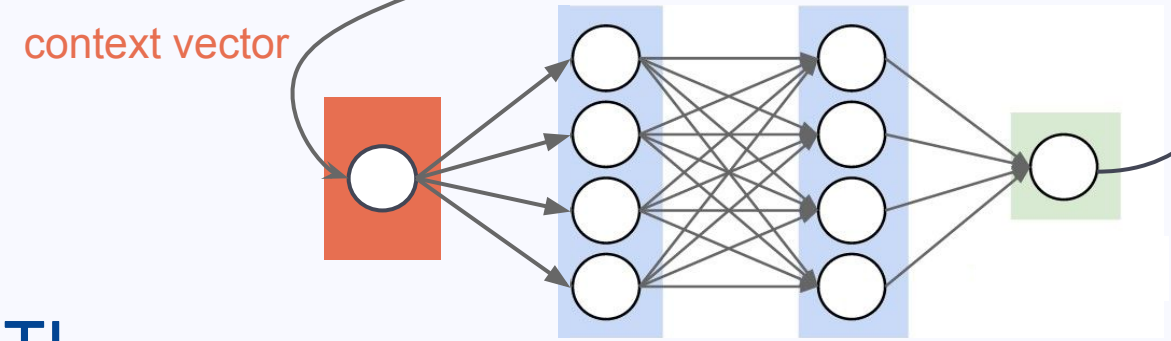

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Encoder-decoder architecture

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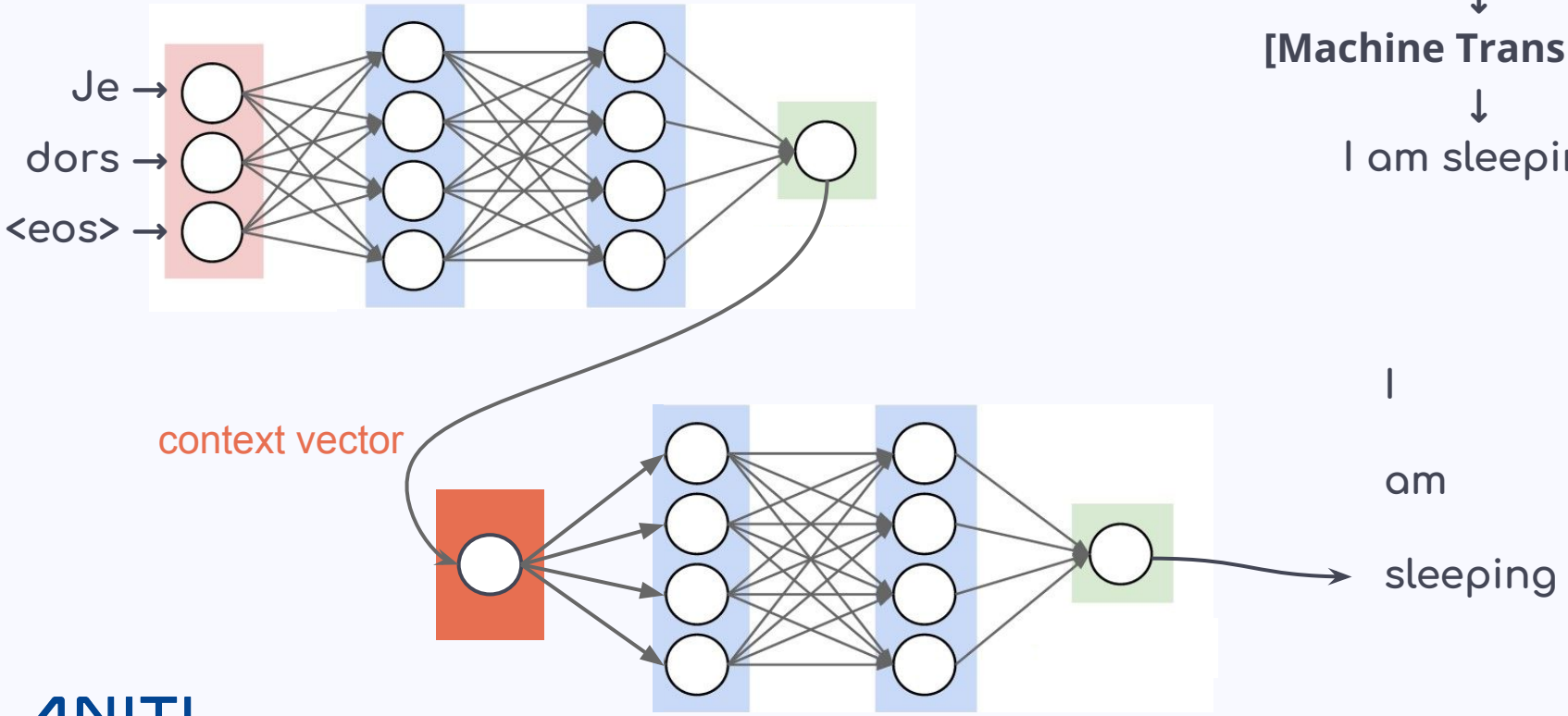
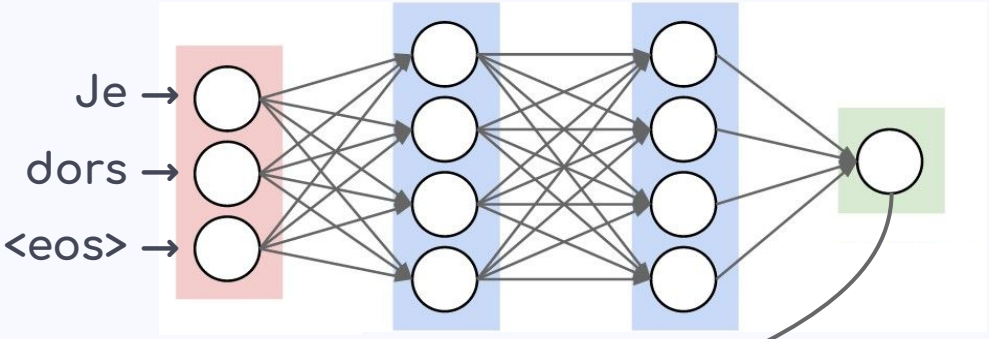
I am sleeping



Encoder-decoder architecture



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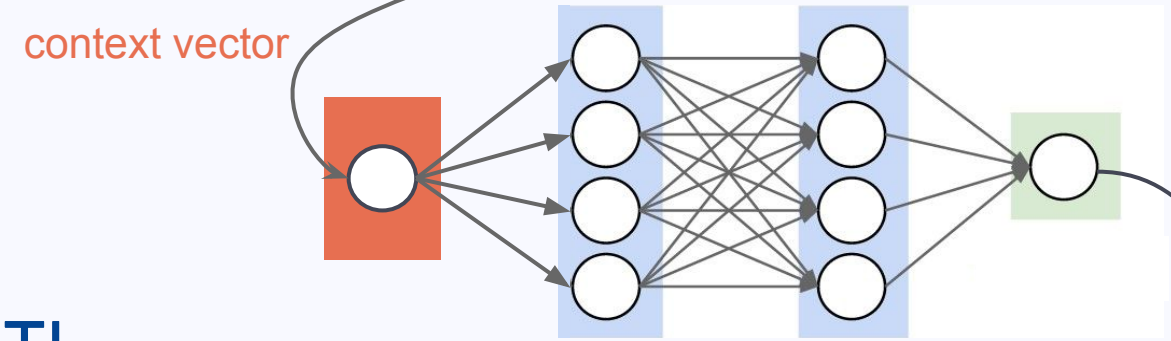


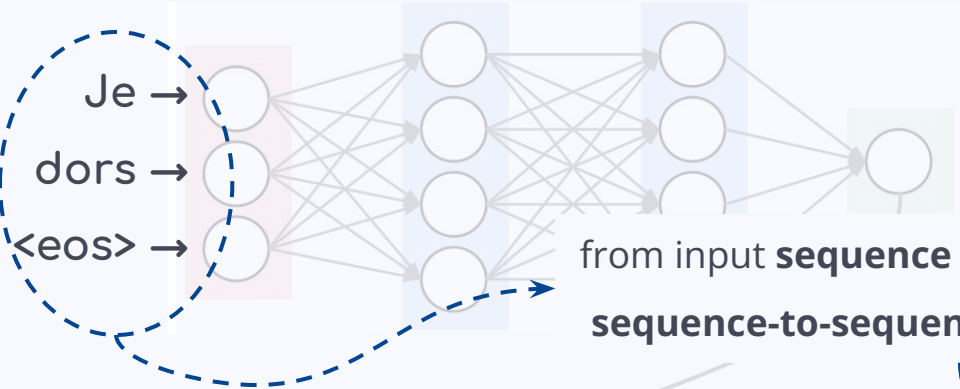
I

am

sleeping

<eos>

Encoder-decoder architecture



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[Machine Translation]

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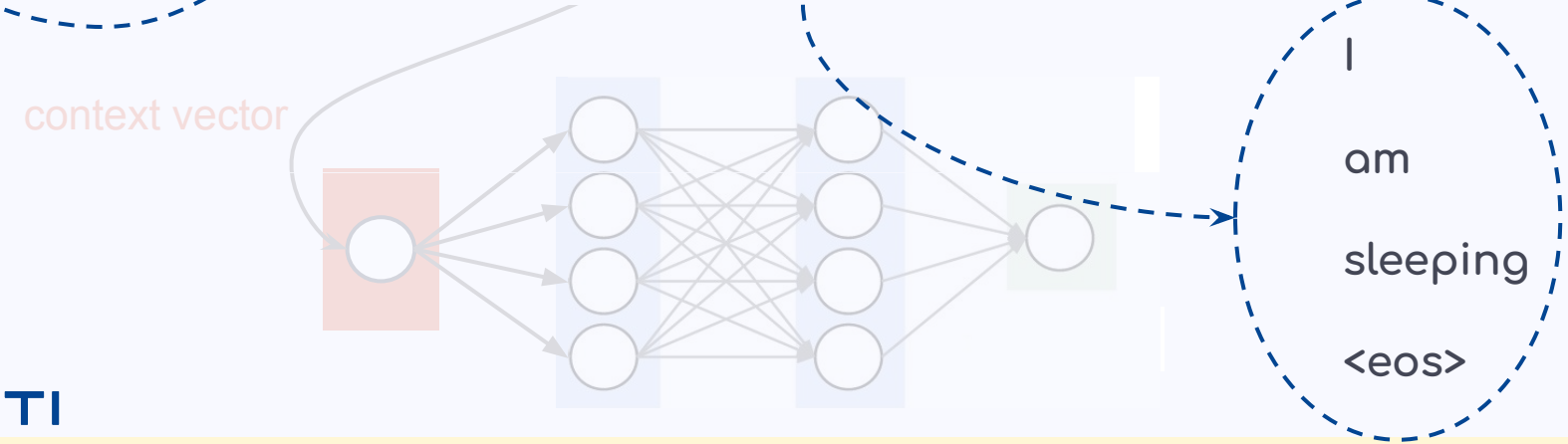
from input **sequence** to output **sequence**...

sequence-to-sequence (seq2seq) model!



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Encoder-decoder: Pros & Cons

+ Better performance, adaptive
+ Generalizes across tasks
+ Adapted to NLP tasks with varying input/output lengths

Encoder-decoder: Pros & Cons

+ Better performance, adaptive
+ Generalizes across tasks
+ Adapted to NLP tasks with varying input/output lengths

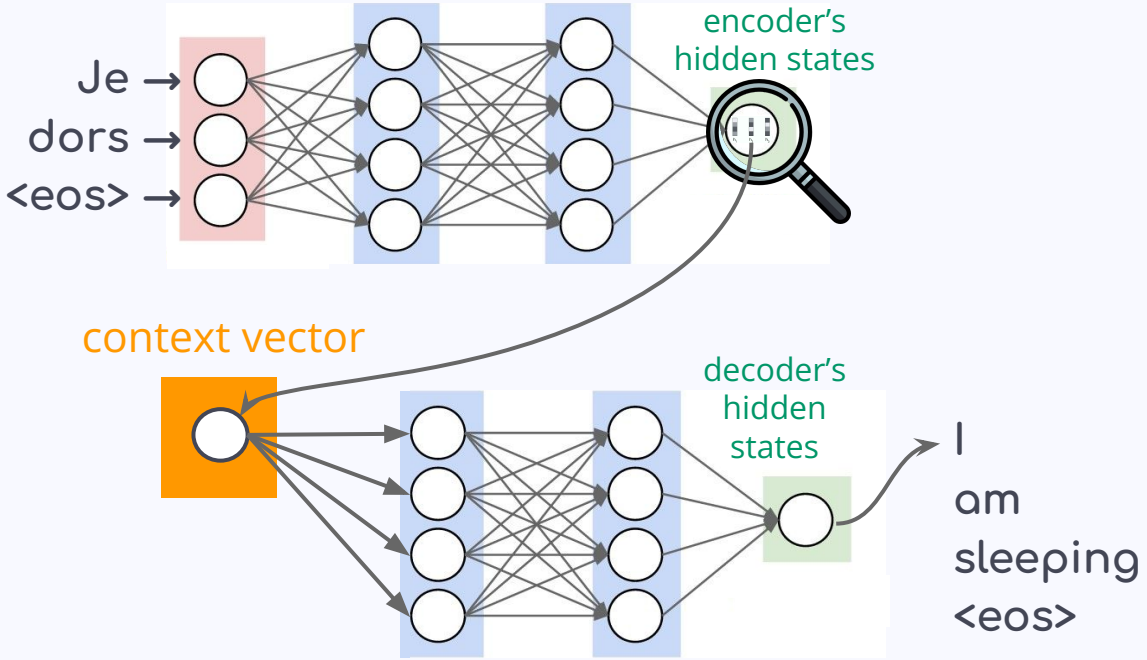
− Lack of explicit alignment!
− Difficulty with long sequences!
− Over-reliance on context vector!

5. Attention mechanism

What is Attention?

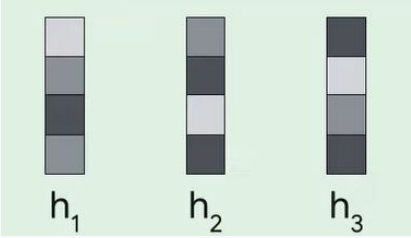
- A **similarity score** between each input token (after passing encoder) and each output token (after passing through decoder)
- **Attention score's** goal: highest scores between related tokens, e.g. *Je* ↔ *I*
- **Attention mechanism**: a NN in-between Encoder-Decoder!

Adding Attention to seq2seq

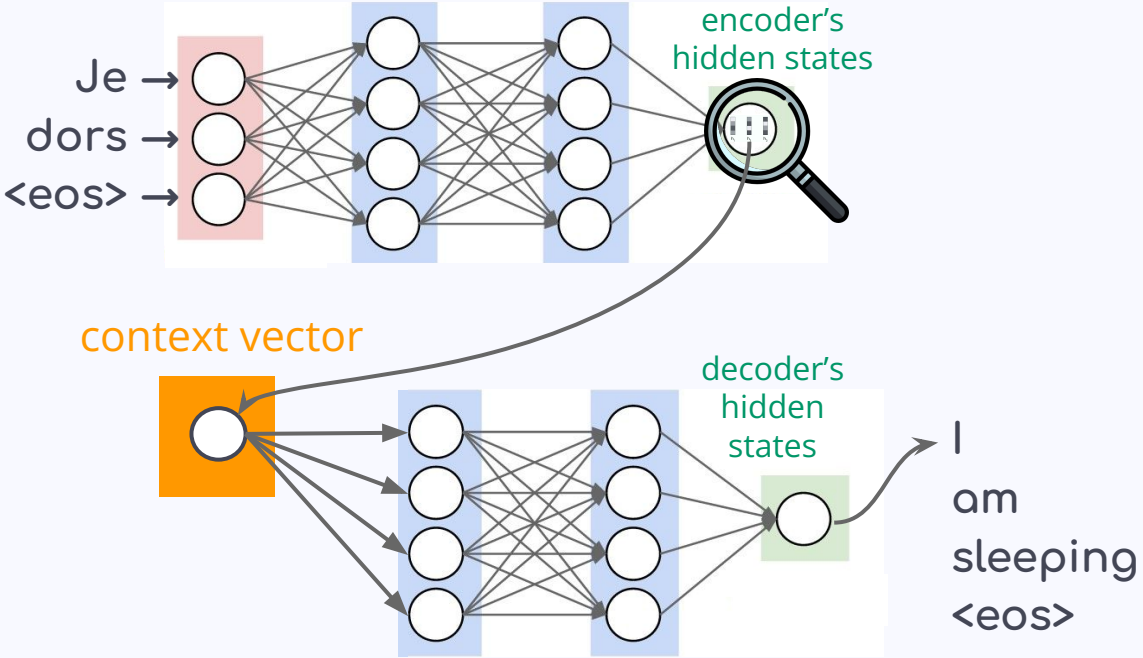


Google Cloud Tech: https://youtu.be/fjIOgb-E41w?si=IdPeyLvbiGDyeaN_

Inside the context vector...

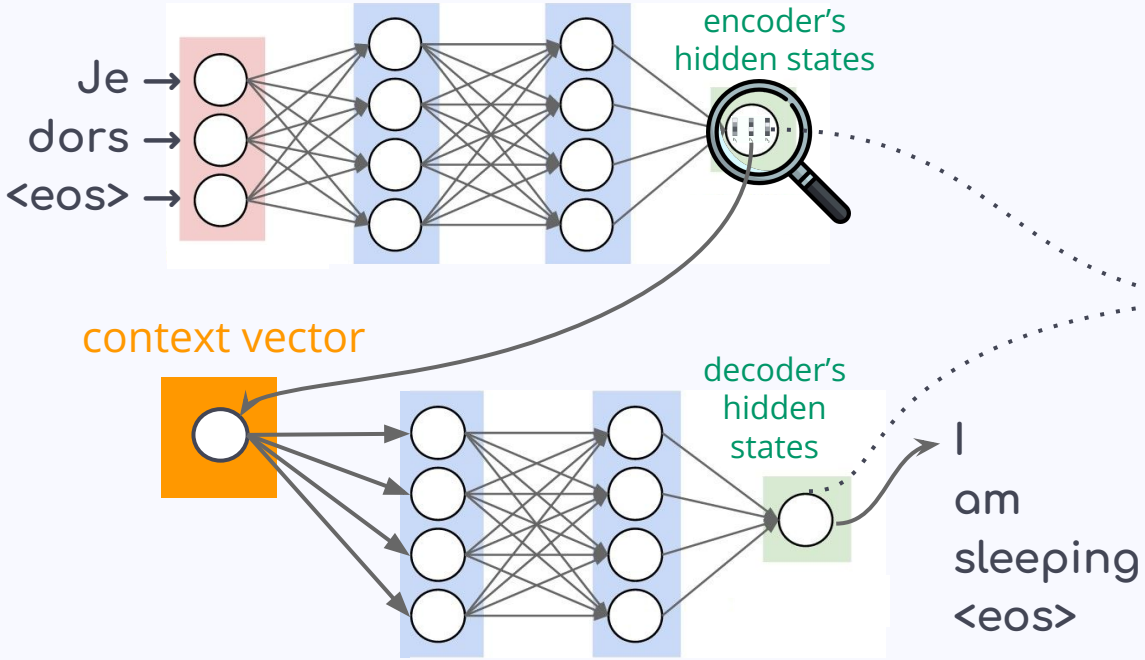


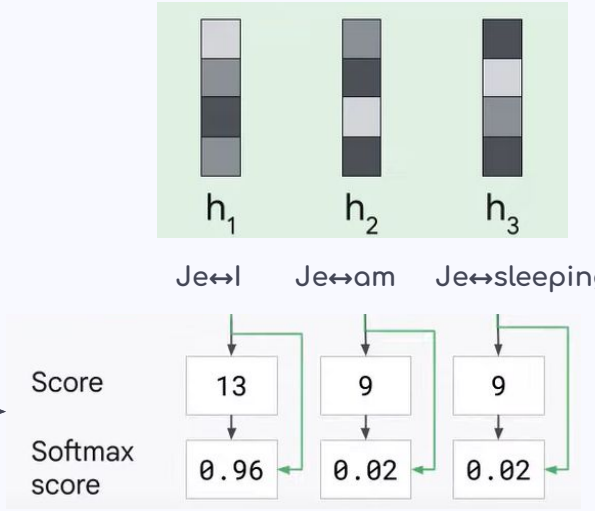
Adding Attention to seq2seq



Google Cloud Tech: https://youtu.be/fjIOgb-E41w?si=IdPeyLvbiGDyeaN_

Adding Attention to seq2seq





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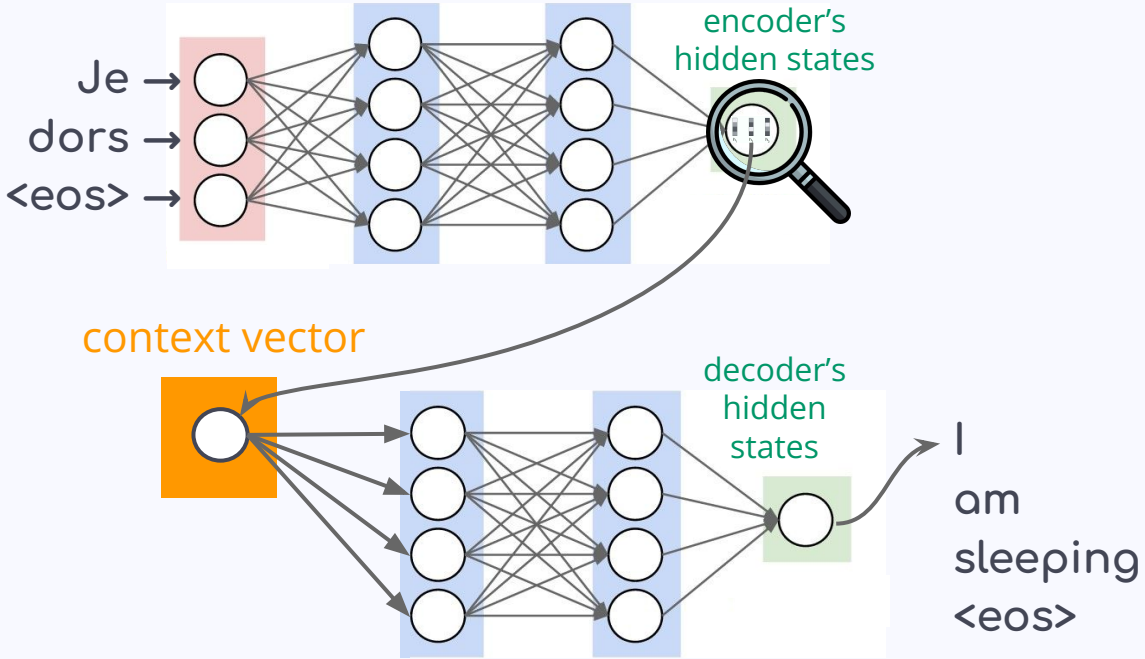
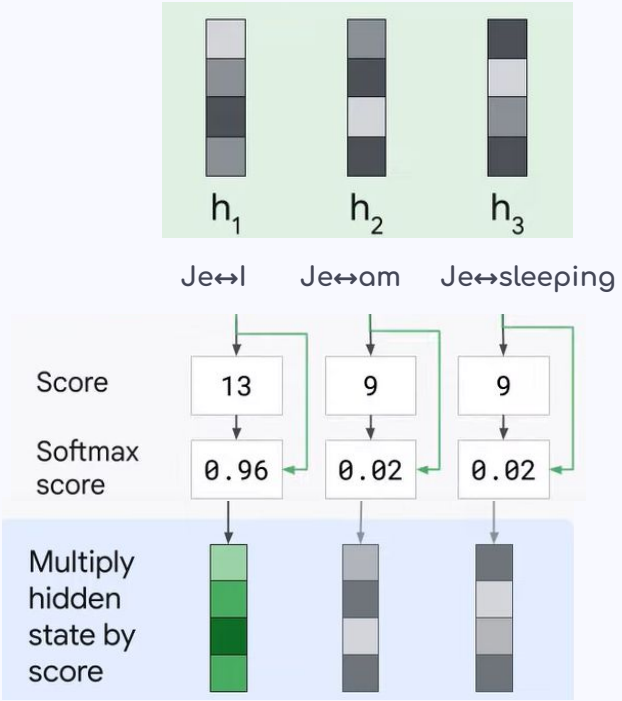
Inside the context vector…

Adding Attention to seq2seq





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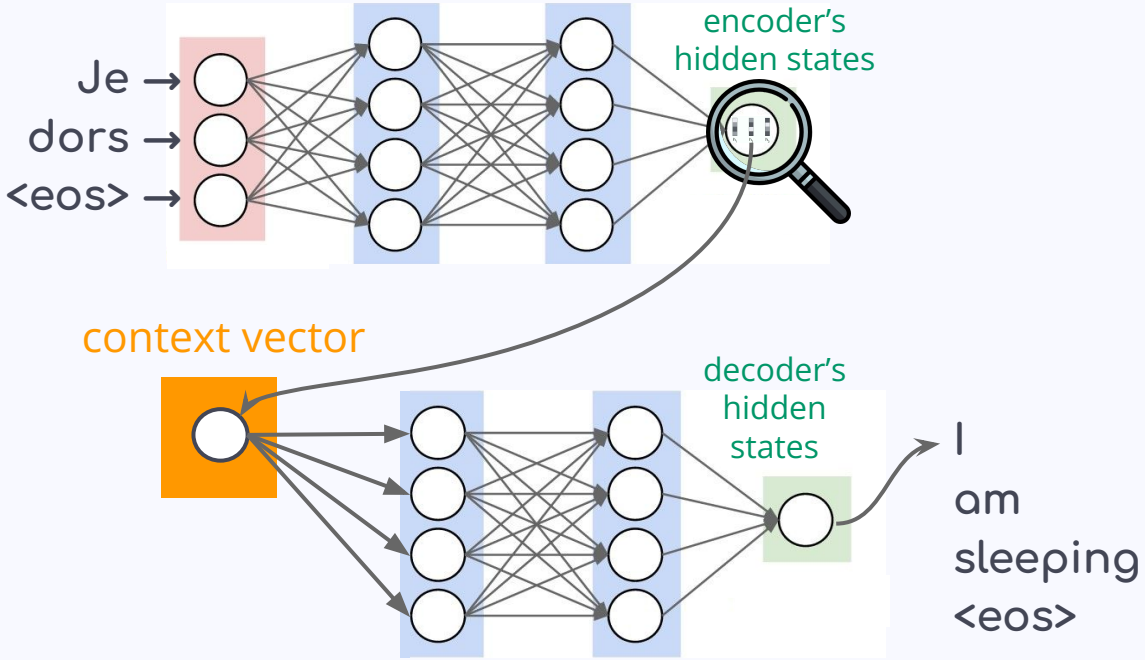
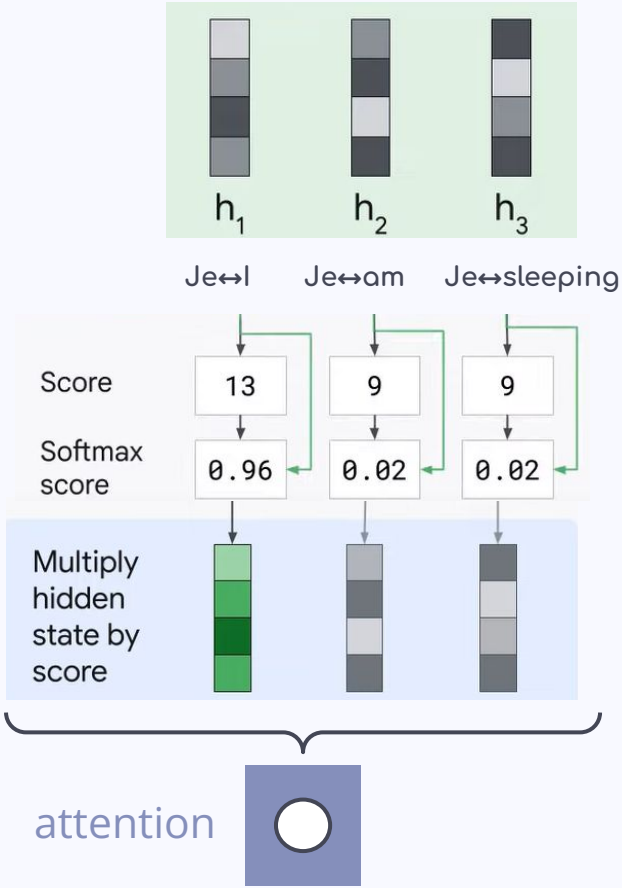
Adding Attention to seq2seq





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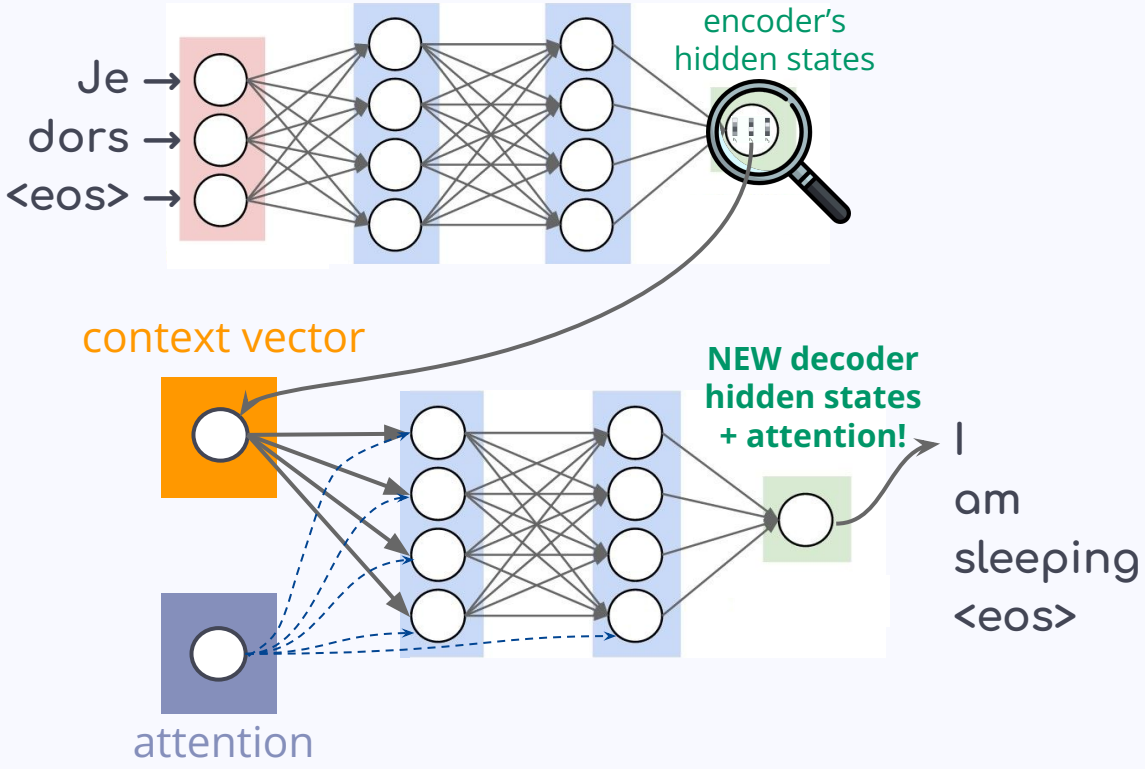
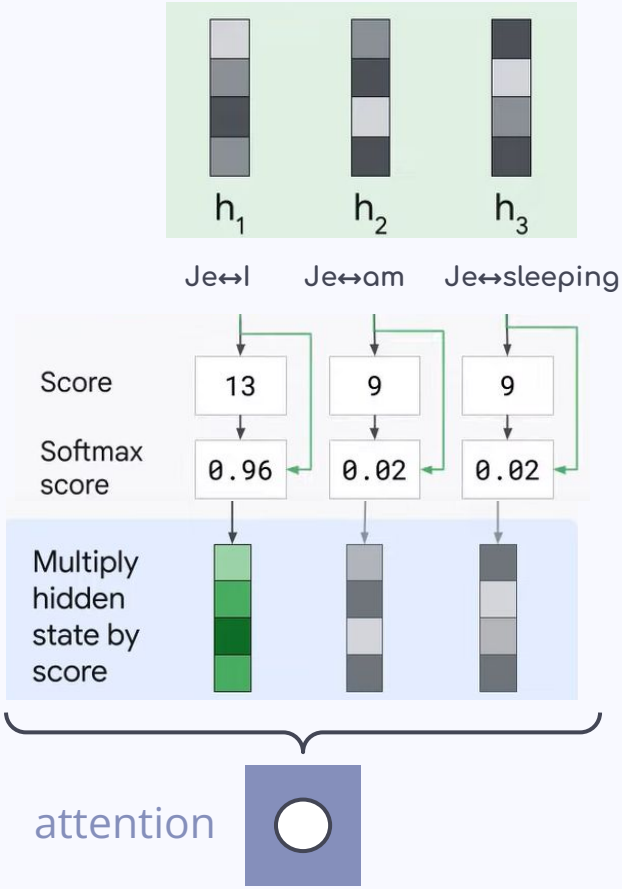
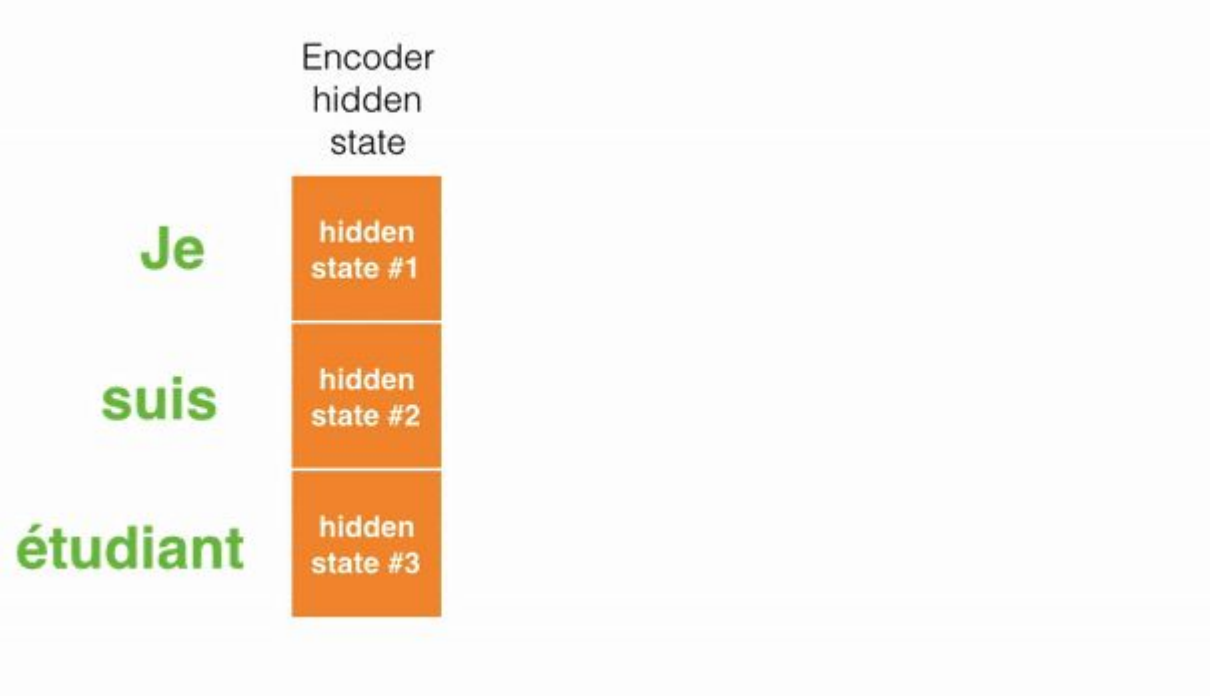
Adding Attention to seq2seq



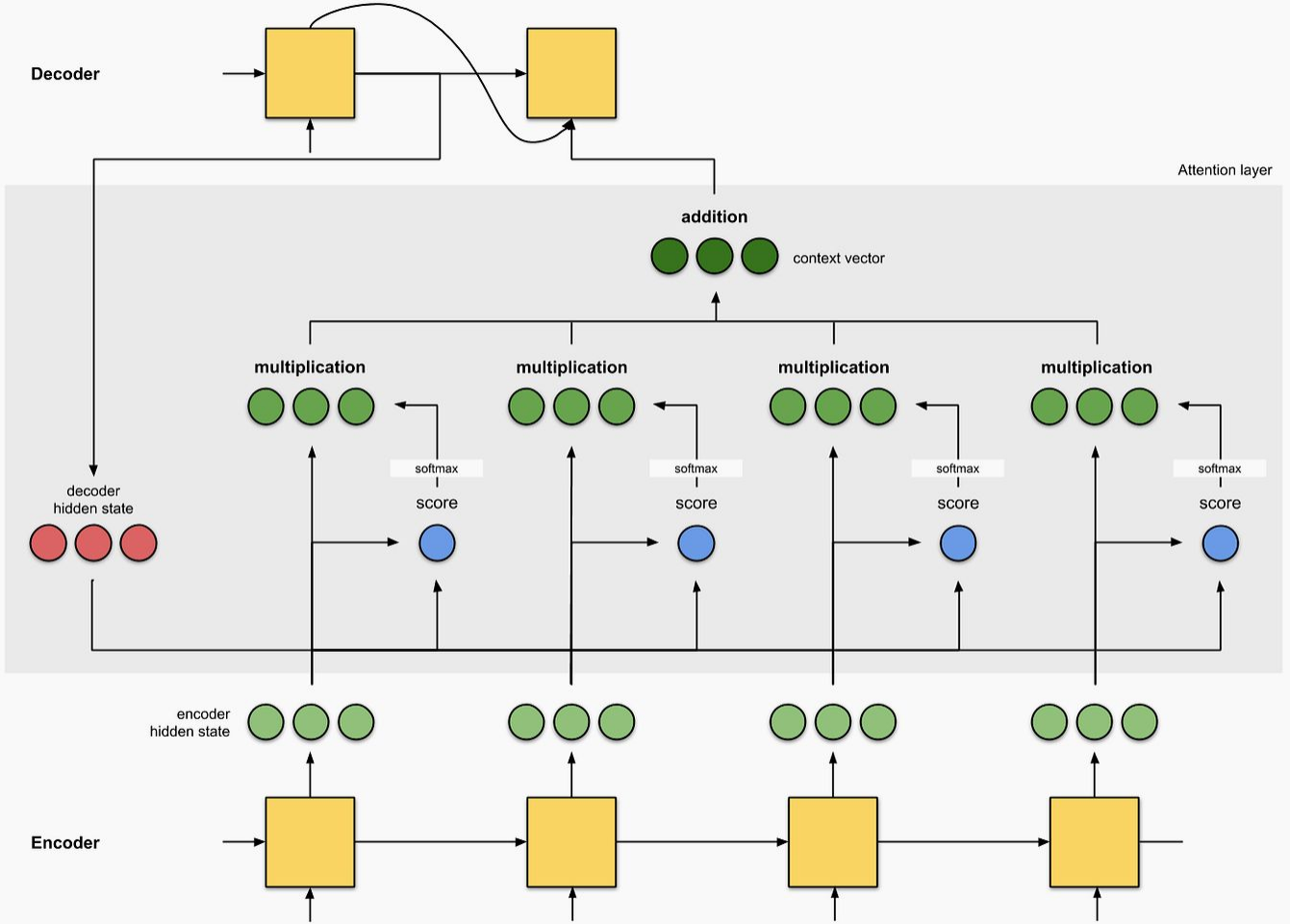


Google Cloud Tech: https://youtu.be/fjIOgb-E41w?si=IdPeyLvbiGDyeaN_

The (ideal) attention weights



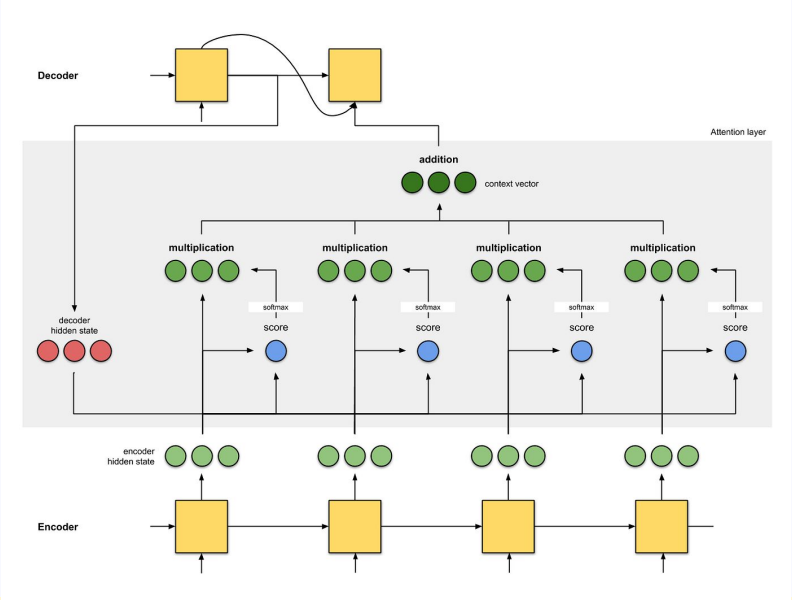
https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/



https://medium.com/data-science/attn-illustrated-attention-5ec4ad276ee3#0458

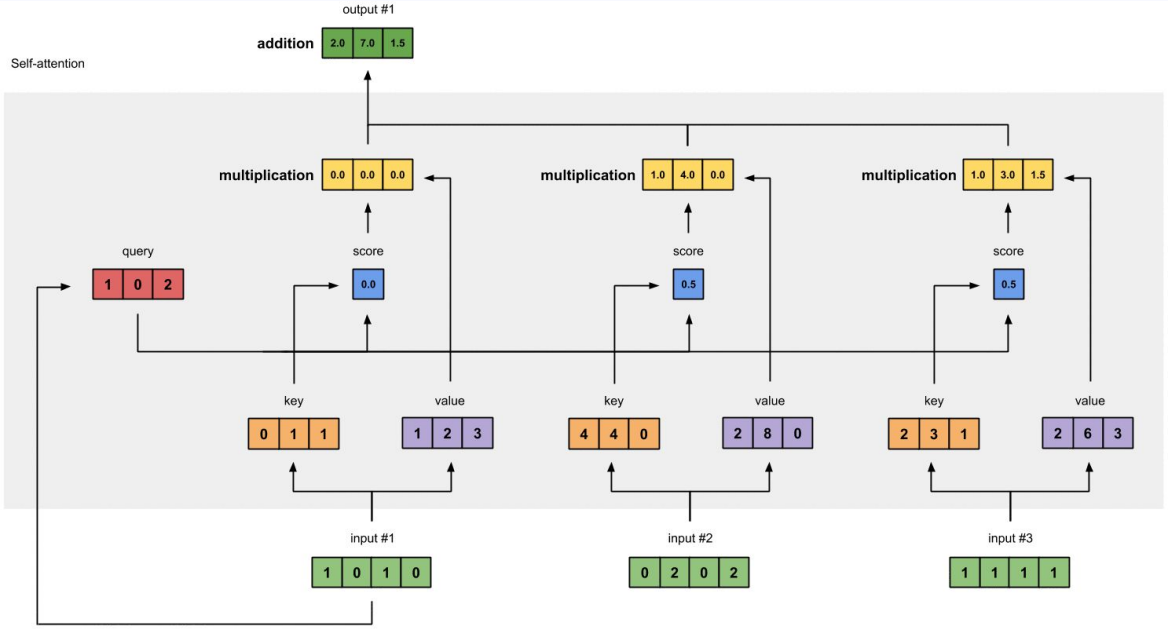
But is it enough attention?

- Vaswani et al. (2017): **Self-attention** = foundation for transformers & LLMs!
- Goodbye seq2seq with attention…



But is it enough attention?

- Vaswani et al. (2017): **Self-attention** = foundation for transformers & LLMs!
- Hello **transformer with self-attention**!



https://medium.com/data-science/illustrated-self-attention-2d627e33b20a

Thank you for your attention (weights)!

Let’s head to Google Colab!

https://colab.research.google.com/drive/1MWqyE46KXhKfL-hpQBMhAHYSlsKejtal?usp=sharing

ANITI

Practical session refresher

- **Padding**: adding 0s to the sentence vector after <eos>, in order to make all our input sequences the same length (0 = ignored)
- **Dropout**: deactivate temporarily some nodes
- **Epoch**: one full training of the NN with all the data
  - **Forward/backward (pass)**: one processing pass of the data in one direction
- **Batch**: how many sentences we will process simultaneously
- **Loss**: function that compares the target and predicted output values, aka how good/bad the predictions are!

Further resources

- A Neural Network Playground: visualize how neural networks look inside!
- 📰 (Reliable) reading material:
  - The Unreasonable Effectiveness of Recurrent Neural Networks
  - Visualizing A Neural Machine Translation Model (Mechanics of Seq2seq Models With Attention) - en français
  - Dive into Deep Learning - everything NN: book, code, math (advanced!)
- ▶️ YouTube tutorials:
  - 3Blue1Brown
  - StatQuest
  - The AI hacker